MATH GRE BOOTCAMP: LECTURE NOTES

IAN COLEY

1. Day 1: Single variable calculus

Topics covered: limits, derivatives, implicit differentiation, related rates, the intermediate value theorem, the mean value theorem, optimisation, L'Hôpital's rule, inverse functions and their derivatives, logarithms and exponential functions and their derivatives.

1.1. Basics.

Definition 1.1. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function. We say that $\lim_{x\to a} f(x) = L$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $|x - a| < \delta$, $|f(x) - L| < \varepsilon$.

We will worry about limits of sequences on another day.

Definition 1.2. We say that $f\colon \mathbb{R} \to \mathbb{R}$ is continuous at $x = a$ if $\lim_{x\to a} f(x) = f(a)$.

Problem 1.3. At what points is the following function continuous?

$$f(x) = \begin{cases} x^2 & x \in \mathbb{Q} \\ x/5 & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

This sort of problem (coupled with having taken real analysis) should remind you that the heuristic 'continuous means you don't pick up your pen' doesn't work in sophisticated situations.

There is also a notion of right continuous and left continuous, where we use only one-sided limits. This is more easily drawn than defined (and they also have no analogue in multivariable calculus, so we will omit the full definition). If this were a blackboard, there would be a better example here.

When are functions discontinuous? Jump discontinuities (almost always in piecewise functions), infinite discontinuities, and removable discontinuities (holes). One could also consider a function discontinuous at the points where it ceases to exist, e.g. $f(x) = \sqrt{x}$ is discontinuous at $x \leq 0$. For examples of continuous functions, think of almost literally any function: polynomials, trigonometric functions, logarithms, exponential functions, etc.

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It is likely unimportant for the GRE, but it might be nice to recall the squeeze theorem just in case:

Theorem 1.4 (Squeeze Theorem)**.** Suppose that $f, g, h \colon \mathbb{R} \to \mathbb{R}$ are three functions. If $f(x) \leq g(x) \leq h(x)$ in a neighbourhood of $x = a$, then

$$\lim_{x \to a} f(x) \leq \lim_{x \to a} g(x) \leq \lim_{x \to a} h(x).$$

This is particularly useful when $f(x)$ and $h(x)$ are continuous at $x = a$ (or one is even constant) but $g(x)$ isn't.

Problem 1.5. Compute $\lim_{x \to 0} x \cdot \sin\left(\frac{1}{x}\right)$.

The second main definition we need for calculus is that of differentiable functions.

Definition 1.6. We say that $f \colon \mathbb{R} \to \mathbb{R}$ is differentiable at $x = a$ if $\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ exists. Alternatively, we can ask that $\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$ exists. We write $f'(a)$ for this value.

Problem 1.7. Prove that these two definitions agree.

Problem 1.8. Prove that if $f(x)$ is differentiable at $x = a$, it is continuous at $x = a$.

When are functions non-differentiable? By the previous exercise, when they're discontinuous. It's important to remember to always check continuity first, as in the following:

Problem 1.9. Describe the set of solutions $(a, b, c) \in \mathbb{R}^3$ such that the following function is continuous and differentiable (everywhere):

$$f(x) = \begin{cases} ax^2 + bx + c & x \leq 1 \\ x \log x & x > 1 \end{cases}$$

Solution. The pair (a, b) determines the equality of derivatives at $x = 1$ and c determines the continuity. To check continuity, we have that $f(1) = a + b + c = 1 \cdot \log(1)$, so that $a+b+c = 0$. Second, by the product rule we have $(x \log x)' = 1 + \log x$, so for differentiability we need $f'(1) = 2a + b = 1 + \log(1) = 1$, so that $2a + b = 1$. We can solve $b = 1 - 2a$, so $a + b + c = a + (1 - 2a) + c = 0$ hence $-a + c = 1$ so $c = 1 + a$ and $b = 1 - 2a$. Thus the set of solutions can be written $(a, 1 - 2a, 1 + a)$.

There are two basic theorems about continuous and differentiable functions we will state now, and a third later.

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Theorem 1.10 (Intermediate Value Theorem)**.** Let $f\colon [a,b] \to \mathbb{R}$ be a continuous function. Assume that $f(a) < f(b)$. Then for any $y \in [f(a), f(b)]$, there exists $c \in [a,b]$ such that $f(c) = y$.

use this to compute what zeroes are

Theorem 1.11 (Mean Value Theorem)**.** Let $f\colon [a,b] \to \mathbb{R}$ be a differentiable function (except at the endpoints). Then there exists a point $c \in [a,b]$ such that

$$f'(c) \cdot (b-a) = f(b) - f(a).$$

I would call these theorems 'fundamental' but we all know that's reserved for a bit later.

1.2. **Derivatives.** Of course we all remember the basic rules for doing derivatives, but let's recall them anyway: let $f, g\colon \mathbb{R} \to \mathbb{R}$ be two functions.

- $(f \pm g)'(x) = f'(x) \pm g'(x)$
- $(c \cdot f)'(x) = c \cdot f'(x)$ for all $c \in \mathbb{R}$
- $(f \cdot g)'(x) = f(x) \cdot g'(x) + f'(x) \cdot g(x)$
- $\left(\dfrac{f}{g}\right)'(x) = \dfrac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{g(x)^2}$ for $g(x) \neq 0$
- $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$

There are a few other functions we should memorise the derivative of:

- $\dfrac{d}{dx} c = 0$ for $c \in \mathbb{R}$
- $\dfrac{d}{dx} x^n = n \cdot x^{n-1}$
- $\dfrac{d}{dx} \sin(x) = \cos(x)$
- $\dfrac{d}{dx} \cos(x) = -\sin(x)$
- $\dfrac{d}{dx} \sinh(x) = \cosh(x)$
- $\dfrac{d}{dx} \cosh(x) = \sinh(x)$

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- $\dfrac{d}{dx}e^x = e^x$
- $\dfrac{d}{dx}\log(x) = \dfrac{1}{x}$

Other trigonometric functions can be computed using the quotient rule.

Perhaps it is also worth remembering the less-used but GRE-noteworthy formula for the second derivative of a function:

Problem 1.12. Prove that

$$f''(x) = \lim_{h\to 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}$$

What other techniques do we use for computing derivatives? Computing the derivatives of inverse functions can be difficult, specifically when we don't have a closed formula for the inverse. What circumstances are those?

Definition 1.13. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function, and suppose that $X \subset \mathbb{R}$ is a set on which f is one-to-one. Then we say that f is invertible on X, and write $f^{-1}(y)$ for the inverse, which is defined by $f^{-1}(y) = x$ if and only if $f(x) = y$.

Problem 1.14. Suppose that $f\colon \mathbb{R} \to \mathbb{R}$ is a function and that for all $x \in X \subset \mathbb{R}$, $f'(x) > 0$. Then f is invertible on X.

Problem 1.15. Suppose that $f\colon \mathbb{R} \to \mathbb{R}$ is a differentiable invertible function. Then $f^{-1}\colon \mathbb{R} \to \mathbb{R}$ is also differentiable, except at those $y = f(x) \in \mathbb{R}$ such that when $f'(x) = 0$.

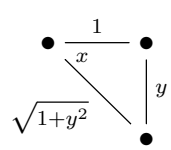
How do we compute the derivative of the inverse? Write $f^{-1} = g$ for simplicity, Then $f(g(y)) = y$, so taking derivatives and using the chain rule we have

$$(f \circ g)'(y) = f'(g(y)) \cdot g'(y) = 1 \implies g'(y) = \frac{1}{f'(g(y))}$$

So as long as we can figure out $g(y)$ and $f'(x)$, we can figure out $g'(y)$.

Problem 1.16. Compute the derivative of $\tan^{-1}(y)$.

Solution. By the formula, we have that the inverse should be the derivative of $\tan(x)$ evaluated on $\tan^{-1}(y)$. The derivative of $\tan(x)$ is $\sec^2(x)$, so we need to figure out $\sec^2(\tan^{-1}(y))$. This is done with a technique I personally call 'draw the triangle'. We know that $\tan^{-1}(y) = x$ for some x, so we need to draw the triangle in which x is one of the angles. We know only that $\tan(x) = y$, so we may draw a right triangle:



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Therefore $\sec^2(x)$ we can compute as the hypotenuse squared over the adjacent side squared, that is $\sec^2(\tan^{-1}(y)) = 1 + y^2$. Therefore the derivative of $\tan^{-1}(y)$ is $\dfrac{1}{1+y^2}$.

This method can be used to compute the derivatives of the other inverse trigonometric and hyperbolic trigonometric functions on the fly, so you don't need to necessarily memorise all of them. That said, you should definitely memorise the above example.

Logarithmic differentiation is a useful technique, and it also recalls implicit differentiation.

Definition 1.17. Let $F\colon \mathbb{R}^2 \to \mathbb{R}$ be a function and let $F(x, y) = c$ implicitly define a function of one variable $f\colon (a, b) \to \mathbb{R}$. Then

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

where F_x, F_y are the partial derivatives of F.

In practice, life is easier than this. We will illustrate with an example.

Problem 1.18. Find the tangent line to the circle $x^2 + y^2 = 25$ at $(3, 4)$.

Solution. In practice, we just take the derivative of everything with respect to x and recall that $y' = dy/dx$. Hence

$$0 = 2x + 2y \cdot y' \implies y' = \frac{-2x}{2y} = -\frac{x}{y}.$$

This is exactly what we get when we use the definition as well. To finish the actual problem, we have

$$\frac{dy}{dx}\Big|_{(x,y)=(3,4)} = -\frac{3}{4}$$

so that the tangent line is $y - 4 = -\dfrac{3}{4}(x - 3)$.

Now, what is logarithmic differentiation? Let $y = f(x)$. Using implicit differentiation (or the chain rule, depending on your perspective), we have

$$(\log y)' = \frac{y'}{y} \implies y' = y \cdot (\log y)'.$$

In the case that $\log y$ is easier to differentiate than y, this is a helpful trick. For example, consider $f(x) = x^x$ or

$$f(x) = \sqrt{\frac{(x+11)^2(x-4)}{(x-1)^2(x+4)}}.$$

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Problem 1.19. Compute the second derivative of $f(x) = x^{x^x}$.

The last type of basic derivative is for parametric functions. Suppose that we define a graph using two functions $x(t)$ and $y(t)$ rather than $y = f(x)$. It's still straightforward to compute the change in y with respect to the change in x:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

This process can be iterated to find the second derivative of y with respect to x, but it's a bit more difficult.

Problem 1.20. Using the quotient rule, compute the formula for $\dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\dfrac{dy}{dx}$.

1.3. **Applications of Derivatives.** Related rates problems, which involve parametric functions as discussed above, show up quite often in the single-variable curriculum. Classic examples include balloons filling up or deflating and basins filling up or emptying of water. Ladders sliding down a wall or shadows lengthening are also common.

Problem 1.21. Suppose we have a right conical coffee filter which is 8cm tall with a radius of 4cm. The water drips through at a constant rate of 2 cubic centimetres per second. When there is one eighth of the original water remaining, how fast is the water level dropping?

Solution. We begin by noting that $V(t) = \dfrac{\pi}{3} \cdot h(t) \cdot r(t)^2$. But because our cone is conical, we also know that the height and radius of the cone are in a fixed ratio: $h(t) = 2 \cdot r(t)$. Because we are solving for $h'(t)$, we will substitute in $r(t) = r(t)/2$. Hence

$$V(t) = \frac{\pi}{3} \cdot h(t) \cdot \frac{h(t)^2}{4} = \frac{\pi}{12} \cdot h(t)^3.$$

Taking the time derivative,

$$V'(t) = \frac{\pi}{4} h(t)^2 \cdot h'(t)$$

Letting $t = t_0$ be the time at which we would like to find the change in height, we know that $V'(t_0) = -2$ no matter what. Therefore we need to find $h(t_0)$ to complete the problem. We know that $V(t_0) = V(0)/8$, and that $V(0) = \dfrac{\pi}{3} \cdot 8 \cdot 4^2 = \dfrac{128\pi}{3}$. So as $V(t_0) = \frac{\pi}{12} \cdot h(t_0)^3 = \frac{16\pi}{3}$, it's pretty clear that $h(t_0) = 4$. Therefore

$$-2 = V'(t) = \frac{\pi}{4} \cdot 4^2 \cdot h'(t_0) \implies -\frac{1}{2\pi} = h'(t_0).$$

We can now turn to optimisation. This is certainly a favourite in the undergraduate curriculum and appears sometimes in the GRE. Why does it work?

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Theorem 1.22 (Extreme Value Theorem)**.** Let $f\colon [a,b] \to \mathbb{R}$ be a continuous function. Then the set $f([a,b])$ has both a maximum value and a minimum value.

Note that it's necessary that $[a,b]$ be a closed interval. The image of an open interval under a continuous function may have neither a maximum nor a minimum, i.e. $f(x) = 1/x$ and the interval $(0,\infty)$.

We can even say more:

Definition 1.23. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a differentiable function. We say that $a \in \mathbb{R}$ is a critical point if $f'(a) = 0$.

We might also include the case that $f\colon \mathbb{R} \to \mathbb{R}$ is a function which is differentiable except at finitely many points call those critical points as well. Their utility is the following theorem, credited to Fermat by various sources.

Theorem 1.24 (Fermat's Boring Theorem)**.** Let $f\colon [a,b] \to \mathbb{R}$ be a differentiable function (except at the endpoints). Then the maxima and minima of $f(x)$ occur at critical points and end points.

There are many different kinds of optimisation problems (as there are related rates problems), but all have the same basic process: we are given both a function to optimise and a constraint. The function to optimise is going to be some $F(x,y)$ in two variables, and the constraint is going to be some equation $g(x,y) = c$. After substituting, we'll have a one-variable problem. Sometimes we will have endpoints, and sometimes it will be implicit that as your one variables gets too big or small that there is no extremum to be found. We then find the critical points and exhaustively check which is the largest and which is the smallest.

Problem 1.25. Suppose we are constructing a window comprised of a semicircle sitting atop a rectangle. Given that the perimeter of the window must be 4 meters, what is the maximum area?

Solution. Let us call x the width and y the height of the rectangle. We know that the area is given by the sum $xy + \frac{\pi}{2} \cdot \left(\frac{x}{2}\right)^2$, the first for the rectangle and the second for the semicircle. Our constraint is $2y + \pi x + x = 4$, the bottom three sides of the rectangle and the arc of the semicircle. It looks like making x our sole variable will be the best path to success, so we will substitute $y = 2 - \frac{1+\pi}{2} \cdot x$. Our function is thus

$$A(x) = x \cdot \left(2 - \frac{1+\pi}{2} \cdot x\right) + \frac{\pi}{2} \cdot \left(\frac{x}{2}\right)^2 = 2x - \frac{4+3\pi}{8} \cdot x^2.$$

Note that this is the equation of a downward-facing parabola, so if x is too big or too small we have $A(x) < 0$. This is obviously a nonsense answer to the question,

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so what we're looking for is the critical point giving the vertex of the parabola – its maximum.

We now need to solve $A'(x) = 2 - \frac{4+3\pi}{4} \cdot x = 0$, so $x = \frac{8}{4+3\pi}$. We can now compute the maximum area:

$$A\left(\frac{8}{4+3\pi}\right) = \frac{8}{4+3\pi} \quad \text{(not a typo).}$$

Problem 1.26. What is the minimum distance between the curve $y = 1/x$ and the origin?

Distance questions seem more difficult, as the optimisation equation we are dealing with is of the form $d(x) = \sqrt{x^2 + f(x)^2}$. But an important observation is that the minimum of $d(x)$ is also achieved by $d(x)^2$, because the distance function is always positive. Moreover, critical points of $d(x)$ and $d(x)^2$ are the same, as

$$\frac{d}{dx}(d(x)^2) = 2 \cdot d(x) \cdot d'(x)$$

and $d(x) > 0$ as long as $x \neq 0$ or $f(x) \neq 0$. The bright side is that $d(x)^2 = x^2 + f(x)^2$ is much easier to differentiate than $d(x)$.

1.4. **Graphical Analysis.** We already know that $f'(x) > 0$ means increasing and $f'(x) < 0$ means decreasing, but now let's recall what the second derivative means. If $f''(x) > 0$, the graph $y = f(x)$ is concave up, so that the graph is open to $+\infty$, and $f''(x) < 0$ is concave down.

Definition 1.27. If $f''(a) = 0$, we say that $x = a$ is a point of inflection.

In the circumstances of optimisation, we can use the second derivative to test whether a critical point is a maximum or minimum.

Theorem 1.28 (Second Derivative Test)**.** Suppose that $x = a$ is a critical point of $f\colon \mathbb{R} \to \mathbb{R}$. Suppose that $f''(x)$ exists in a neighbourhood of a. If $f''(a) < 0$, then $f(a)$ is a local maximum. If $f''(a) > 0$, then $f(a)$ is a local minimum. If $f''(a) = 0$, the test is inconclusive.

There is also the first derivative test: suppose $f'(x)$ exists and is continuous near $x = a$. If $f'(x) < 0$ to the left of a and $f'(x) > 0$ to the right of a, then $f(a)$ is a local minimum. If $f'(x) > 0$ to the left of a and $f'(x) < 0$ to the right of a, then $f(a)$ is a local maximum. This can often be more useful for optimisation problems when the second derivative is too difficult to compute. However, as in the above example of the window, we see that $A''(x) = -\frac{3\pi}{4}$ constantly, so any extreme values that exist must be maxima.

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Problem 1.29. Suppose that $y = f(x)$ is smooth (i.e. has continuous derivatives of all orders) and that $f(1) = 2$ is a local maximum. Order the values $f(1)$, $f'(1)$, $f''(1)$.

Solution. We know that $f(1) = 2$, so that settles that. Because $x = 1$ is a local extremum, we must have $f'(1) = 0$. Finally, because this extreme point is a maximum, we must be concave down so $f''(1) < 0$. Hence $f''(1) < f'(1) < f(1)$.

1.5. **L'Hôpital's Rule.** The last topic worth remembering L'Hôpital's rule, which comes surprisingly in the second quarter of calculus at UCLA but we can recall now.

Theorem 1.30. Let $f, g\colon \mathbb{R} \to \mathbb{R}$ be two functions and $a \in \mathbb{R} \cup \{\pm\infty\}$. If $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = c$ where $c \in \{0, \pm\infty\}$, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

Expressions of the form $\frac{0}{0}$ and $\pm\frac{\infty}{\infty}$ are called indeterminate forms. L'Hôpital's rule can also be used to solve problems which are not immediately in an indeterminate form. The easiest case is expressions of the form $f(x) \cdot g(x)$ which give rise to the indeterminate form $0 \cdot \infty$. By rearranging to

$$\frac{f(x)}{1/g(x)} \text{ or } \frac{g(x)}{1/f(x)}$$

we will obtain an actual indeterminate form. Which one to choose depends on whether $1/f(x)$ or $1/g(x)$ is easier to differentiate.

Problem 1.31. Compute the following limit:

$$\lim_{x\to 2} \frac{1}{x-2} - \frac{2x+1}{x^2-4}$$

Solution. If you plug in $x = 2$, we don't obtain an indeterminate form, but we obtain something that looks like $\infty - \infty$. This is our clue to combine the fractions into an indeterminate form. With a common denominator,

$$\frac{x+2}{x^2-4} - \frac{2x}{x^2-4} = \frac{-x+2}{x^2-4} = \frac{-(x-2)}{(x-2)(x+2)} = \frac{-1}{x+2}.$$

In this case we don't even need to use L'Hôpital's rule to finish the problem since we had some nice cancellation. In other cases we might not be so lucky.

Problem 1.32. Compute the following limit:

$$\lim_{x\to\infty} x^{1/x}$$

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Solution. These problems are also related to L'Hôpital's rule as well. Plugging in, we obtain ∞^0. We notice that if we took the log of this expression, $\log\left(x^{1/x}\right) = \frac{1}{x} \cdot \log x$ yields the form $0 \cdot \infty$. We can now rearrange it to $\dfrac{\log x}{x}$ and finish the problem:

$$\lim_{x\to\infty} \log\left(x^{1/x}\right) = \lim_{x\to\infty} \frac{\log x}{x} \overset{\text{L'H}}{=} \lim_{x\to\infty} \frac{1}{x} = 0.$$

But of course this solves the wrong question. If we say $L = \lim_{x\to\infty} x^{1/x}$, then $\log L = 0$ (as log is a continuous function so commutes with limits). Thus $L = 1$.

This same type of solution works if we have the form 1^∞, as $\log(1^\infty) = \infty \cdot \log 1$ yields $\infty \cdot 0$. This concludes the differential side of single-variable calculus.

2. Day 2: Single variable calculus

Topics covered: the integral, area between curves, volumes of revolution, the fundamental theorem of calculus, u-substitution, integration by parts, trigonometric integration, partial fractions, arc length and surface area, sequences and series, convergence tests, Taylor polynomials and power series, root and ratio tests.

2.1. Integrals.

Let us start with the definition of the Riemann integral. To do so, we need to recall the limit of a sequence.

Definition 2.1. Let $\{x_n\}_{n\in\mathbb{N}}$ be a sequence. We say that $\lim_{n\to\infty} x_n = L$ if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that whenever $n > N$, $|x_n - L| < \varepsilon$.

The difference here is that there is no function floating around. We will revisit the intricacies of sequences and series later in this lecture.

Definition 2.2. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function. We define the left Riemann integral of f on the interval $[a, b]$ to be the limit

$$\lim_{n\to\infty} \frac{b-a}{n} \cdot \sum_{i=0}^{n-1} f\left(a + \frac{b-a}{n} \cdot i\right)$$

We define the right Riemann integral of f on the interval $[a, b]$ to be the limit

$$\lim_{n\to\infty} \frac{b-a}{n} \cdot \sum_{i=1}^{n} f\left(a + \frac{b-a}{n} \cdot i\right)$$

We say that the Riemann integral of f on the interval $[a, b]$ if the above exist and agree. In that case, we write

$$\int_a^b f(x)\, dx$$

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There is more we could say on this definition, but it's not necessary for the GRE.

We also call the individual terms of these limits the left and right Riemann sums, denoted $L_n f$ or $R_n f$. There are a few other approximations for integrals, including the midpoint and trapezoid approximations. The trapezoid approximation is the average of the left and right, and the midpoint rule uses $\left(a + \frac{b-a}{2n} \cdot (2i+1)\right)$ in its argument.

We will almost never use the right and left Riemann sums, as these limits are not calculable in practice, but it's important to know a few things. If a function is increasing, then the left Riemann sum will always underestimate the actual value of the integral, and the right Riemann sum will always overestimate it.

The Riemann integral is not guaranteed to exist for an arbitrary function, but it must exist for our favourite functions.

Proposition 2.3. If $f\colon [a,b] \to \mathbb{R}$ is bounded and continuous at all but finitely many points, then $\int_a^b f(x)\,dx$ exists.

We could assign this as an exercise, but it's a bit difficult and not necessary at all for the GRE. This isn't a necessary and sufficient condition, but it is certainly good enough for almost all purposes.

The integral is linear, just as the derivative was. In particular, this means that

$$\int_a^b c \cdot f(x)\,dx = c \cdot \int_a^b f(x)\,dx$$

and

$$\int_a^b f(x) \pm g(x)\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$$

Moreover, recalling the definition via Riemann sums, we can always split an integral into intermediate chunks. For any $c \in [a,b]$, we have

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

By definition we will say $\int_a^a f(x)\,dx = 0$, so using the additivity above

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

This way we can make sense of integrals where our interval $[a,b]$ happens to be oriented the wrong way (i.e. $a > b$).

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In practice, we will always compute the integral using the Fundamental Theorem of Calculus.

Theorem 2.4 (FTC I)**.** Assume that $f\colon [a,b] \to \mathbb{R}$ is a continuous function. If $F\colon [a,b] \to \mathbb{R}$ is a function such that $F'(x) = f(x)$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Hence the calculation of integrals will amount to the calculation of antiderivatives (i.e. the function $F(x)$ above).

There's a companion theorem that we state now.

Theorem 2.5 (FTC II)**.** Let $f\colon [a,b] \to \mathbb{R}$ be a continuous function, and define $F(x) = \displaystyle\int_a^x f(t)\,dt$. Then $F'(x) = f(x)$.

We can extend this theorem using the chain rule:

Problem 2.6. Prove that

$$\frac{d}{dx}\int_{a(x)}^{b(x)} f(t)\,dt = f(b(x))\cdot b'(x) - f(a(x))\cdot a'(x).$$

One last thing to say at this point is the definition of an improper integral. Suppose that we are trying to integrate $f(x)$ on an unbounded region, say $[0,\infty)$, or over a region $[a,b]$ on which $g(x)$ has an infinite discontinuity (say at $x = a$). Then we can define the integral (should it exist) as a limit of integrals as defined above, e.g.

$$\int_0^\infty f(x)\,dx := \lim_{R\to\infty}\int_0^R f(x)\,dx \qquad \int_a^b g(x)\,dx := \lim_{h\to 0^+}\int_{a+h}^b g(x)\,dx$$

These limits are not guaranteed to exist. The most common types of integrals we consider in this situation are functions $f(x)$ such that $\lim\limits_{x\to\infty} f(x) = 0$ so that the integral at least has a chance of converging. We will readdress this issue in the section on series.

2.2. **Applications of the integral.** But before we get into techniques, why bother computing integrals at all? For one, the integral of $f(x)$ on $[a,b]$ gives the signed area of the region under the graph of $f(x)$. This can even be used in reverse: integrals can be calculated using geometry.

Problem 2.7. Compute the integral

$$\int_{-2}^{2} \sqrt{4-x^2}\,dx$$

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Solution. The equation $y = \sqrt{4 - x^2}$ corresponds to $x^2 + y^2 = 4$, i.e. a circle of radius 2. Our graph is the top half of this circle. Therefore the area under the curve is half the area of the circle, 2π. This problem can be solved otherwise, but it's much more annoying.

In the same way, we can compute the area between curves with integrals. We need only to compute the area under the upper curve and subtract the area under the lower curve. The main question in situations like this is which curve is on top and which is on bottom.

Problem 2.8. Compute the area between the curves $y = \sqrt{x}$ and $y = x^2$ in the first quadrant.

Solution. Thinking on the graphs, we can recall that $\sqrt{x}$ is above x^2 in the region $[0, 1]$ where these curves intersect. Therefore the integral we need to compute is

$$\int_0^1 \sqrt{x} - x^2 \, dx.$$

Luckily, we can get antiderivatives of these functions very easily:

$$\int_0^1 \sqrt{x} - x^2 \, dx = \frac{2x^{3/2}}{3} - \frac{x^3}{3}\Big|_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}.$$

The other application is to surfaces and volumes of revolution. Supposing that we rotate a curve $y = f(x)$ around the x-axis, the height of the curve becomes the radius of a disc, which we then need to integrate along the interval $[a, b]$ in question, whatever it is:

$$V = \pi \int_a^b f(x)^2 \, dx$$

Problem 2.9. What is the volume of the region created by rotating $y = \log x$ around the x-axis between $x = 1$ and $x = e^2$?

It's harder to compute the volume when we rotate $y = f(x)$ around the y-axis. Rather than use the method of discs, we use the method of cylindrical shells. In this case, we are computing the area of a cylinder of radius r and height h, which is given by $2\pi \cdot r \cdot h$. In our case, the radius is x and the height is $f(x)$, hence

$$V = 2\pi \int_a^b x \cdot f(x) \, dx$$

Problem 2.10. Compute the volume of the region created by rotating $y = 1 - 2x + 3x^2 - 2x^3$ from $[0, 1]$ around the y-axis.

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Arc length of a curve is another application. If we are looking at the infinitesimal change in the length of a curve, it travels dx in the x direction and dy in the y direction. Therefore its total length is $ds = \sqrt{(dx)^2 + (dy)^2}$. In the case that $y = f(x)$, we have

$$s = \int ds = \int \sqrt{(dx)^2 + \left(\frac{dy}{dx}\right)^2} = \int_a^b \sqrt{1 + f'(x)^2}\, dx$$

Problem 2.11. Compute the arc length of $y = \cosh x$ over the interval $[0, 2]$.

Sometimes we also want to calculate the surface areas of regions of revolution, not just the volumes. In this case, we again have a formula: the infinitesimal amount of surface area is given by the arc length along the surface times the circumference of the disc, which in the case of rotation around the x-axis is $2\pi f(x)$. Thus

$$S = 2\pi \int f(x)\sqrt{1 + f'(x)^2}\, dx$$

Problem 2.12. Compute the surface area of a sphere of radius R, using the curve $y = \sqrt{R^2 - x^2}$.

2.3. **Integration techniques.** Now besides guessing at antiderivatives, what are our other integration techniques? The first is u-substitution, which is is our answer to the chain rule.

Theorem 2.13 (u-substitution)**.** Suppose that $h(x)$ is a continuous function and we can write $h(x) = f'(g(x))g'(x)$. Then

$$\int_a^b h(x)\, dx = \int_{g(a)}^{g(b)} f(u)\, du.$$

Problem 2.14. Evaluate the integral of $f(x) = 2x \cdot \sin(x^2)$ on $[0, \sqrt{\pi/2}]$.

Solution. We see that $g(x) = x^2$ and $f(u) = \sin u$ is a good choice. Hence

$$\int_0^{\sqrt{\pi/2}} 2x \cdot \sin(x^2)\, dx = \int_0^{\pi/2} \sin u\, du = -\cos u\Big|_0^{\pi/2} = 1.$$

The second is integration by parts, which is our answer to the product rule. It steps from the following observation: if we consider the product $(f \cdot g)' = f' \cdot g + f \cdot g'$ and integrate both sides, we obtain $f \cdot g = \int g\, df + \int f\, dg$.

Theorem 2.15 (Integration by Parts)**.** Suppose that $f(x)$ has the form $u(x) \cdot v'(x)$. Then

$$\int f(x)\, dx = \int u(x) \cdot v'(x)\, dx = u(x) \cdot v(x) - \int v(x) \cdot u'(x)\, dx.$$

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This is useful when your function is a product of a part which is easy to integrate and a part which is difficult to integrate. Another situation is when part of your function will differentiate to zero and the other part will not differentiate, as in $x \cdot e^x$. A general mnemonic for what functions should be chosen for $v'(x)$ is LIATE: logarithms, inverse trigonometric functions, algebraic (e.g. polynomials), trigonometric functions, and finally exponential functions. Note that these latter two types in particular are almost never ideal.

Problem 2.16. Compute the integral of $f(x) = \log x$.

Solution. If we follow our mnemonic, we will choose $dv = \log x \, dx$. This means that $u = 1$. Therefore $v = \dfrac{1}{x} \, dx$ and $du = dx$. Hence:

$$\int \log x \, dx = x \log x - \int x \cdot \frac{1}{x} \, dx = x \log x - \int 1 \, dx = x \log x - x.$$

It doesn't seem like it would work, and then it does.

Problem 2.17. Compute the integral of $f(x) = x^2 \cdot e^x$.

The next method is the method of trigonometric substitution. There are some integrals that do not lend themselves to either of the above methods, but require a special kind of u-substitution. We recognise this situation in the case that one of the Pythagorean identities holds, usually the following:

Problem 2.18. Compute the integral of $f(x) = \sqrt{1 + x^2}$.

Solution. The trigonometric identity we are looking for in this situation is that $1 + \tan^2 \theta = \sec^2 \theta$. Therefore if we substitute $x = \tan \theta$, we will only need to integrate $\sec \theta$, which is much more tractable than the current issue.

However, we have to consider the integration term. If $x = \tan \theta$, then $dx = \sec^2 \theta \, d\theta$. Thus

$$\int \sqrt{1 + x^2} \, dx = \int \sec \theta \cdot \sec^2 \theta \, d\theta.$$

We leave it in this form because we can now use integration by parts to solve this problem: let $dv = \sec^2 \theta \, d\theta$ and $u = \sec \theta$. Then $v = \tan \theta$ and $du = \sec \theta \tan \theta$, so

$$\int \sec^3 \theta \, d\theta = \sec \theta \cdot \tan \theta - \int \sec \theta \tan^2 \theta \, d\theta.$$

We now use that $\tan^2 = \sec^2 \theta - 1$ as we did above so that

$$\int \sec^3 \theta \, d\theta = \sec \theta \tan \theta - \int \sec \theta (\sec^2 \theta - 1) \, d\theta$$

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which rearranged gives us

$$2\int \sec^3\theta\,d\theta = \sec\theta\tan\theta - \int \sec\theta\,d\theta$$

Therefore the last thing to compute is $\int \sec\theta\,d\theta$. This is a hard calculation that one must memorise. The key is that we can apply u-substitution if we decide (as if by magic) to multiply by $\dfrac{\sec\theta+\tan\theta}{\sec\theta+\tan\theta}$. I will leave it to the reader to verify

$$\int \sec\theta\,d\theta = \log(\tan\theta+\sec\theta).$$

Putting this all together completes the problem, sort of. Though we know that

$$\int \sec^3\theta\,d\theta = \frac{\sec\theta\tan\theta}{2} - \frac{1}{2}\log(\tan\theta+\sec\theta)$$

we were asked a question about $f(x)$, not $f(\theta)$. We know that $x=\tan\theta$, so we can make that substitution. In order to determine what $\sec\theta$ is in terms of x, we draw the triangle as in yesterday's lecture. The Pythagorean theorem will tell us that $\sec\theta=\sqrt{1+x^2}$. If we then put it all together,

$$\int \sqrt{1+x^2}\,dx = \frac{x\sqrt{1+x^2}}{2} - \frac{1}{2}\log(x+\sqrt{1+x^2}).$$

The final integration technique is the method of partial fractions. We are only capable with our techniques to integrate fairly simple rational functions, so the method of partial fractions allows us to break up complicated expressions into integrable ones. The idea is that any quotient $\dfrac{p(x)}{q(x)}$ of polynomials comes from a sum of polynomials whose denominators are the irreducible factors of $q(x)$. We can integrate expressions of the form $\dfrac{a}{x-r}$ or $\dfrac{ax+b}{x^2+r}$.

Problem 2.19. Compute the integral of $\dfrac{4}{x^4-1}$.

Solution. We factor the denominator as $(x+1)(x-1)(x^2+1)$, and so set up

$$\frac{4}{x^4-1} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{Cx+D}{x^2+1}$$

Multiplying through by the denominator,

$$4 = A(x-1)(x^2+1) + B(x+1)(x^2+1) + (Cx+D)(x+1)(x-1)$$

We can plug in particular values for x to compute A and B. If $x=-1$,

$$4 = A(-2)(2) \implies A=-1$$

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If $x = 1$,

$$4 = B(2)(2) \implies B = 1$$

For the last two variables, we need to do some brute multiplication. Omitting the details,

$$4 = Cx^3 + (2 + D)x^2 - Cx + (2 - D)$$

which implies that $C = 0$ and $D = -2$. Hence

$$\frac{4}{x^4 - 1} = \frac{-1}{x + 1} + \frac{1}{x - 1} + \frac{-2}{x^2 + 1}$$

which we can now integrate:

$$\begin{aligned}\int \frac{4}{x^4 - 1}\, dx &= \int \frac{-1}{x + 1}\, dx + \int \frac{1}{x - 1}\, dx + \int \frac{-2}{x^2 + 1} dx \\ &= -\log|x + 1| + \log|x - 1| - 2\tan^{-1}(x)\end{aligned}$$

There's a bit of a complication if the irreducible factor of $q(x)$ is not exactly of the form $x^2 + r$, but completing the square and some further manipulation can always get us to that form.

Problem 2.20. Prove that

$$\int \frac{dx}{x^2 + r} = \frac{1}{r}\tan^{-1}\left(\frac{x}{\sqrt{r}}\right)$$

2.4. **Sequences and series.** We gave above the definition of a sequence and the definition of a convergent sequence. We can now recall what a series is.

Definition 2.21. A series is a sequence $\{S_n\}_{n\in\mathbb{N}}$ defined by a sequence $\{a_i\}_{i\in\mathbb{N}}$ such that $S_n = \sum_{i=0}^{n} a_i$. A series converges if the sequence $\{S_n\}$ converges as above. We write $\sum_{i=0}^{\infty} a_i$ for $\lim_{n\to\infty} S_n$.

Sometimes we will call $\{a_i\}$ the series and leave the fact that we are taking sums implicit.

The simplest type of sequence/series that we encounter is the geometric series, which is defined by $a_i = a_0 \cdot r^i$ for some $r \in \mathbb{R}$. In this circumstance, there is an easy criterion for when the series $\{a_i\}$ converges.

Problem 2.22. Prove that the series $\{a_i = a_0 \cdot r^i\}$ converges if and only if $|r| < 1$.

In this situation, the infinite sum has the formula $\dfrac{a_0}{1 - r}$.

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2.5. **Convergence tests.** Unfortunately, most series aren't geometric, so we have convergence tests to decide whether they converge. If a series does not converge, we say it diverges. The first test is the following easy check:

Theorem 2.23 (Divergence Test)**.** The series $\{a_n\}$ diverges if $\lim_{n\to\infty} a_n \neq 0$.

Theorem 2.24 (p-test)**.** Suppose we have the series $a_n = \dfrac{1}{n^p}$ for some $p \in \mathbb{R}$. The series $\{a_n\}$ converges if and only if $p > 1$.

This situation is not so common, but is a useful tool (as we will soon see).

Theorem 2.25 (Limit Comparison Test)**.** Suppose that $\{a_n\}, \{b_n\}$ are nonnegative series. Consider the quantity

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L$$

where $L \in [0, \infty]$. Then we have the following three conclusions:

- If $L = 0$, the series $\{a_n\}$ converges if $\{b_n\}$ converges
- If $L = \infty$, the series $\{b_n\}$ converges if $\{a_n\}$ converges
- If $L \in (0, \infty)$, the series $\{a_n\}$ converges if and only if $\{b_n\}$ converges

It is sometimes useful to bear in mind the contrapositives of these statements. Therefore while many series do not take the form of the p-test, they can be limit-compared to a 'p-series' and thus we can determine their convergence. These two also combine to answer the standing question about improper integrals:

Theorem 2.26 (Integral Test)**.** Suppose that $\{a_n\}$ is a positive series such that there exists a continuous function $f(x)$ satisfying $f(n) = a_n$. Then the series $\{a_n\}$ converges if and only if the improper integral $\displaystyle\int_0^\infty f(x)\,dx$ exists (converges).

Therefore the same types of tests that check for the convergence of infinite series can be used to check the convergence of improper integrals when the limits in question are too difficult to compute. Here, however, is one case that can be computed directly.

Problem 2.27. Show that $\displaystyle\int_0^1 \frac{1}{x^p}\,dx$ converges if and only if $p < 1$.

Problem 2.28. When does the integral $\displaystyle\int_0^\infty \frac{1}{x^p}$ converge?

Thus far we have spoken of positive series, but there is specific test for alternating series that is occasionally useful:

Theorem 2.29 (Alternating Series Test)**.** Suppose that $\{a_n\}$ is a series satisfying:

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- The series is alternating, i.e. a_i and a_{i+1} have different signs for every $i \in \mathbb{N}$
- The series does not pass the divergence test, i.e. $\lim_{n\to\infty} a_n = 0$
- The series is decreasing, i.e. $|a_i| > |a_{i+1}|$ for every $i \in \mathbb{N}$.

Then the series converges.

For example, we saw above that the series $\left\{\frac{1}{n}\right\}$ does not converge (the harmonic series), but the alternating series $\left\{\frac{(-1)^n}{n}\right\}$ does. As a remark, the factor $(-1)^n$ is the easiest way to see that a series is alternating, but $\cos(\pi \cdot n)$ does the job as well.

2.6. **Taylor polynomials and series.** Before getting into the last two tests, we need to recall the definition of a Taylor polynomial and a Taylor series. To begin, we can define a Taylor polynomial and then explain its utility.

Definition 2.30. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function of class C^k. The kth Taylor polynomial of f centred at $x = a$ is defined by

$$T_k f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2} + \cdots + \frac{f^{(k)}(a)(x-a)^k}{k!}$$

This is the best polynomial approximation to $f(x)$ with the property that the first k derivatives at $x = a$ agree. For example, the first Taylor polynomial is just the tangent line to $x = a$, i.e. the linear approximation. If $a = 0$, then usually we use the name Maclaurin instead of Taylor.

The error of a Taylor polynomial can be approximated using what would be the next term. In particular,

$$|T_k f(b) - f(b)| \leq \frac{M \cdot |b-a|^{k+1}}{(k+1)!}$$

where M is the maximum of $|f^{(k+1)}(x)|$ on the interval $[a, b]$ or $[b, a]$ (whichever makes sense). In nice circumstances, M takes its maximum at a or b and so a more complicated calculation is not necessary.

Problem 2.31. Compute $\sqrt{101}$ using the third Taylor polynomial of an appropriate function. What is the approximate error?

Supposing that f is smooth, we can take this definition to infinity.

Definition 2.32. The Taylor series of $f(x)$ centred at $x = a$ is the infinite sum

$$T(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)(x-a)^k}{k!}$$

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In ideal circumstances, we have $T(x) = f(x)$ for all $x \in \mathbb{R}$, but this is not guaranteed. This infinite sum may not even converge for some values of x, or for most values of x for that matter. This is where our last two tests come in handy.

Theorem 2.33 (Ratio Test)**.** Consider a series $\{a_n\}$ which may or may not be positive, and consider the limit

$$\rho = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$$

Then $\{a_n\}$ converges absolutely if $\rho < 1$, diverges if $\rho > 1$, and is inconclusive if $\rho = 1$.

Recall that a series is said to converge absolutely if $\{|a_n|\}$ converges.

In the case that the series depends on a parameter x, this gives us a function $\rho(x)$ that we demand be ≤ 1 to have a chance at convergence. Those x for which $\rho(x) < 1$ is called the radius of convergence. If we also determine whether the cases $\rho(x) = 1$ converge (using other techniques), we obtain the interval of convergence.

Problem 2.34. Compute the interval of convergence of

$$f(x) = \sum_{n=1}^{\infty} \frac{2x^n}{n^2}$$

There is another test that is sometimes more helpful (though rarely).

Theorem 2.35 (Root Test)**.** Consider a series $\{a_n\}$ which may or may not be positive, and consider the limit

$$\rho = \lim_{n\to\infty} |a_n|^{1/n}$$

Then $\{a_n\}$ converges absolutely if $\rho < 1$, diverges if $\rho > 1$, and is inconclusive if $\rho = 1$.

Both these tests are checking to what extent the series we are considering is geometric with common ratio ρ. Hence series that are mostly polynomial (and not geometric) will yield an inconclusive root/ratio test. One should attempt to apply a p-test to these (perhaps via a limit comparison). Series that include factorial or exponential terms are the target for the root and ratio tests.

We end by recalling some useful Taylor expansions and their radius of convergence.

- $e^x = \displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n!}, \quad x \in \mathbb{R}$
- $\dfrac{1}{1-x} = \displaystyle\sum_{n=1}^{\infty} x^n, \quad |x| < 1$

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- $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad x \in \mathbb{R}$
- $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad x \in \mathbb{R}$

We can use these building blocks to construct other Taylor series using substitution, differentiation, and integration.

Problem 2.36. Compute the Taylor series for $\tan^{-1}(x)$ centred at $a = 0$. What is its radius of convergence?

Solution. This looks like a dubious prospect, but we notice that

$$\frac{d}{dx}\tan^{-1}(x) = \frac{1}{1+x^2}$$

and the righthand side looks a lot like a Taylor series we already know. In particular,

$$\frac{1}{1-u} = \sum_{n=0}^{\infty} u^n \implies \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty}(-x^2)^n = \sum_{n=0}^{\infty}(-1)^n x^{2n}$$

We now need to integrate this Taylor series to obtain the one for $\tan^{-1}(x)$. We do so term-by-term,

$$\tan^{-1}(x) = \sum_{n=0}^{\infty}\int(-1)^n x^{2n}\,dx = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2n+1}$$

but need to recall that we might need to add a constant term. The constant term of the Taylor series is $\tan^{-1}(0) = 0$, so we don't need to add anything in this case

What happens to the radius of convergence? Integrating or differentiating doesn't do anything, but substituting does. We know that the series converges for $|u| < 1$, but now $u = -x^2$,so

$$|-x^2| < 1 \implies |x|^2 < 1 \implies |x| < \sqrt{1} = 1$$

In general substitutions will change the radius of convergence, but in this case it happens to stay the same.

3. Day 3: Multivariable calculus

Topics covered: basics on vectors in 3 dimensions, planes, parametric equations, arc length and speed, limits and continuity in multiple variables, partial derivatives, differentiability and tangent planes, gradient and directional derivatives, multivariable chain rule, optimisation.

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3.1. **Vectors in $\mathbb{R}^3$.** Since multivariable calculus takes place with two variables (at least in general and at most for our purposes), graphs will occur in $2 + 1 = 3$ dimensions. Thus we should learn a little bit about $\mathbb{R}^3$. In particular, we need to familiarise ourselves with vector operations (that will reappear in generality for our linear algebra section).

A vector in $\mathbb{R}^3$ is a triple $\langle x, y, z\rangle$ which we think of as an arrow from $(0, 0, 0)$ to (x, y, z). Between any two points in $\mathbb{R}^3$ we can obtain another vector, namely

$$(a_1, a_2, a_3) \to (b_1, b_2, b_3) \sim \langle b_1 - a_1, b_2 - a_2, b_3 - a_3\rangle$$

We'll need to recall the two main operations on vectors in $\mathbb{R}^3$. The first is common to every vector space, which is the dot product (or scalar product):

$$\langle a_1, a_2, a_3\rangle \cdot \langle b_1, b_2, b_3\rangle = a_1b_1 + a_2b_2 + a_3b_3$$

The dot product is linear in each argument, that is,

$$(c\vec{v}) \cdot \vec{w} = c(\vec{v} \cdot \vec{w}) = \vec{v} \cdot (c\vec{w})$$

and

$$(\vec{u} + \vec{v}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w}$$

It's also commutative, which is pretty obvious from its definition. We also define the norm of a vector using this:

$$\|\vec{v}\|^2 = \vec{v} \cdot \vec{v}$$

We say that $\vec{v}$ and $\vec{w}$ are orthogonal if $\vec{v} \cdot \vec{w} = 0$.

The second is the cross product, which only exists on $\mathbb{R}^3$ (at least in this form):

$$\langle a_1, a_2, a_3\rangle \times \langle b_1, b_2, b_3\rangle = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$$

where $\hat{i} = \langle 1, 0, 0\rangle$, $\hat{j} = \langle 0, 1, 0\rangle$, and $\hat{k} = \langle 0, 0, 1\rangle$ are the three unit basis vectors in $\mathbb{R}^3$. The cross product is linear in each variable as well, but it is anticommutative:

$$\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$$

The cross product is actually defined by a universal property: it is a bilinear operation such that $\vec{v} \times \vec{w}$ is orthogonal to both $\vec{v}$ and $\vec{w}$, that the ordered set $\{\vec{v}, \vec{w}, \vec{v} \times \vec{w}\}$ obeys the right-hand rule, and that

$$\|\vec{v} \times \vec{w}\| = \|\vec{v}\| \cdot \|\vec{w}\| \cdot \sin\theta$$

where θ is the planar angle between the two vectors. In particular, $\vec{v} \times \vec{w}$ gives a normal vector to the plane spanned by $\vec{v}$ and $\vec{w}$. But before we investigate that, let's look at planes in general:

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There are two main ways to define a plane (in $\mathbb{R}^3$ at least). A plane is the solutions to a linear function in $\mathbb{R}^3$, so the equation looks like

$$ax + by + cz = d$$

for fixed $a, b, c, d \in \mathbb{R}$. The better way to think about it is to consider a plane as a set of vectors that are orthogonal to the normal vector to the plane:

$$\vec{n} \cdot \vec{v} = 0$$

But this defines a plane that passes through the origin. To move the plane to a point elsewhere, we move it by a fixed amount:

$$\vec{n} \cdot (\vec{v} - \vec{v}_0) = 0$$

But now $\vec{n} \cdot \vec{v}_0 = d$ for some $d \in \mathbb{R}$, so we obtain

$$\vec{n} \cdot \vec{v} = d$$

and letting $\vec{n} = \langle a, b, c \rangle$ and $\vec{v} = \langle x, y, z \rangle$ recovers the above formula.

The cross product is particularly convenient when solving the following problems:

Problem 3.1. Find the equation of the plane in $\mathbb{R}^3$ passing through $P = (0, 0, 1)$, $Q = (1, 0, 0)$ and $R = (1, 1, 1)$.

Solution. Any three non-colinear points defines a unique plane in $\mathbb{R}^3$, and we can take for granted that these points are non-colinear (or check quickly). Recall above we said that if we know that a plane is spanned by two vectors $\vec{v}$ and $\vec{w}$, then $\vec{v} \times \vec{w}$ is normal to the plane. If we know three points, we can come up with two vectors:

$$\vec{v} = \vec{PQ} = \langle 1, 0, -1 \rangle, \quad \vec{w} = \vec{PR} = \langle 1, 1, 0 \rangle$$

The cross product is

$$\vec{n} = \vec{w} \times \vec{w} = \langle 1, -1, 1 \rangle$$

Thus the equation of the plane is $x - y + z = d$, where d is some constant. We can compute it by plugging in any point that is already on the plane, say $(0, 0, 1)$. Hence $d = 0 - 0 + 1 = 1$, so the plane has the equation $x - y + z = 1$.

3.2. **Parametric curves.** We now need to look at parametrised curves in $\mathbb{R}^3$. These are functions $\vec{r}\colon \mathbb{R} \to \mathbb{R}^3$ which we write as $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$. Skipping over some details, limits, continuity, and differentiability of these parametric functions is determined precisely by the component functions.

But what is the meaning of the derivative in this case? If it's done componentwise, then $r'(t) = \langle x'(t), y'(t), z'(t) \rangle$, so every time we plug in a time t we get a whole vector in $\mathbb{R}^3$ instead of a number. How are we supposed to use this to find the tangent line? Luckily, we can shed the chains of $\mathbb{R}^2$ and define lines parametrically instead. The

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most appropriate form of the linear approximation to a curve at $x = a$ in $\mathbb{R}^2$ looks like

$$L(x) = f(a) + f'(a)(x - a)$$

We can do something similar. We can define a parametric curve in $\mathbb{R}^3$ by

$$\begin{aligned}\vec{\ell}(t) &= \langle x(t_0) + x'(t_0)(t - t_0), y(t_0) + y'(t_0)(t - t_0), z(t_0) + z'(t_0)(t - t_0)\rangle \\ &= \vec{r}(t_0) + \vec{r}\,'(t_0) \cdot (t - t_0)\end{aligned}$$

So what do we have? We have the point $\vec{r}(t_0)$ at which we are taking this linear approximation (and this is a line), and we have a slope $\vec{r}\,'(t_0)$ which determines the line's direction.

Problem 3.2. Find the linear approximation to the curve $\vec{r}(t) = \langle t^2, t^3, 2t - 1\rangle$ at time $t = 2$.

What about arc length? We went over how to do this in $\mathbb{R}^2$ earlier: the arc length of the curve $y = f(x)$ is

$$\int_a^b \sqrt{1 + f'(x)^2}\, dx$$

which we obtained by trying to integrate $ds = \sqrt{(dx)^2 + (dy)^2}$. Now, we aren't going to want to integrate with respect to x, because these curves are functions of time t. Moreover, we have three components, so that

$$ds = \sqrt{(dx)^2 + (dy)^2 + (dz)^2}$$

This is the length of the diagonal of the infinitesimal cube with sides dx, dy, dz. Thus by 'factoring out' a dt from all these terms,

$$\begin{aligned}\int ds &= \int \sqrt{(dx)^2 + (dy)^2 + (dz)^2} \\ &= \int \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt \\ &\int \|\vec{r}\,'(t)\|\, dt\end{aligned}$$

No surprise: we are integrating the length of the velocity, i.e. the (directionless) speed of the curve at every point. This is, in fact, the same formula as we were dealing with before. The planar curve $y = f(x)$ can be parametrised as $\langle t, f(t)\rangle$, which has derivative $\langle 1, f'(t)\rangle$ and thus speed $\sqrt{1 + f'(t)^2}$. Thus we can forget the old formula and stick with the new.

Problem 3.3. Compute the arc length of the helix $\vec{r}(t) = \langle \sin t, -\cos t, t\rangle$ from $t = 0$ to $t = 2\pi$.

In Progress

One may recall studying curvature or other horrible topics, but these don't seem to appear on the GRE so we will not revisit them.

3.3. **Multivariable functions.** We will give everything in terms of two variables for now, but the same could be done for 3 or n variables without changing definitions very much.

Suppose now that $f\colon \mathbb{R}^2 \to \mathbb{R}$ is a function. The definition of a limit is still the same.

Definition 3.4. We say that $\lim_{(x,y)\to(a,b)} f(x,y) = L$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $\|(x,y)-(a,b)\| < \delta$, $\|f(x,y) - L\| < \varepsilon$.

Now talk about how to try to compute limits, and approaching along different paths to show that limits don't exist

Problem 3.5. Prove that if $\lim\limits_{(x,y)\to(a,b)} f(x,y) = L$, then for all $\gamma\colon [0,1] \to \mathbb{R}$ such that $\gamma(0) = (a,b)$, then $\lim\limits_{t\to 0} f(\gamma(t)) = L$.

Problem 3.6. Prove that $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy}{x^2+y^2}$ does not exist. Hint: find two paths that give different limits.

Problem 3.7. Prove that $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy^2}{x^2+y^2} = 0$.

Solution. For this, we will recall polar coordinates, which we will use more tomorrow. If we convert the point (x,y) into polar coordinates, then we are instead taking the limit $r \to 0$ and under the substitution $x = r\cos\theta$, $y = r\sin\theta$, we need to solve

$$\lim_{r\to 0} \frac{r^3 \cos\theta \sin^2\theta}{r^2} = \lim_{r\to 0} r(\cos\theta\sin^2\theta)$$

But now we can apply the squeeze theorem to the problem:

$$0 \le \lim_{r\to 0} |r(\cos\theta\sin^2\theta)| \le \lim_{r\to 0} |r| = 0$$

and conclude that the middle limit must be zero as well.

In general, these types of limits exist when the numerator is a higher degree than the denominator and do not otherwise. In the case that we believe the limit to exist, polar coordinates and the squeeze theorem is usually the way to prove it.

Continuity is defined the same way using these limits. A function is continuous if the limit at all points in its domain is the actual value of the function. All functions from single variable calculus are still continuous in multivariable calculus, except that now we allow both x and y to appear.

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3.4. **Partial derivatives.** Differentiability is defined slightly differently for multivariable functions. Instead of having one derivative, we have several.

Definition 3.8. The partial derivative of $f\colon \mathbb{R}^2 \to \mathbb{R}$ at (a, b) with respect to x is (if it exists) the limit

$$\partial_x f(a,b) = \lim_{h\to 0} \frac{f(a+h,b) - f(a,b)}{h}$$

The partial derivative with respect to y is

$$\partial_y f(a,b) = \lim_{k\to 0} \frac{f(a,b+k) - f(a,b)}{k}$$

Definition 3.9. We say that $f\colon \mathbb{R}^2 \to \mathbb{R}$ is differentiable at (a, b) if both partial derivatives $\partial_x f(a, b)$ and $\partial_y f(a, b)$ exist *and* $f(x, y)$ is locally linear at (a, b).

The specific definition does not matter too much, but we have a nice property.

Proposition 3.10. If $\partial_x f(a, b)$ and $\partial_y f(a, b)$ exist and are continuous in a neighbourhood of (a, b), then $f(x, y)$ is differentiable at (a, b).

In this case, we can define the tangent plane to $f(x, y)$ at (a, b) and it is actually the linear approximation: the tangent plane is spanned by the partial derivative in the x-direction and the one in the y-direction. We didn't go over above how to parametrise a plane using two variables, but we can do so now:

$$P(s,t) = s \cdot \langle 1, 0, \partial_x f(a,b)\rangle + t \cdot \langle 0, 1, \partial_y f(a,b)\rangle + \langle a, b, f(a,b)\rangle$$

But this isn't particularly helpful for us, since we would like a form in terms of (x, y, z). Instead, we will define the tangent plane using its normal vector. Because we know two vectors in the plane, which moreover are linearly independent, we take their cross product:

$$\begin{aligned}\langle 1, 0, \partial_x f(a,b)\rangle \times \langle 0, 1, \partial_y f(a,b)\rangle &= \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & \partial_x f(a,b) \\ 0 & 1 & \partial_y f(a,b) \end{bmatrix} \\ &= \langle -\partial_x f(a,b), -\partial_y f(a,b), 1\rangle\end{aligned}$$

Hence our equation is $\langle -\partial_x f(a,b), -\partial_y f(a,b), 1\rangle \cdot \langle x-a, y-b, z-f(a,b)\rangle = 0$. If we work this out and rearrange it, it becomes

$$z = \partial_x f(a,b)(x-a) + \partial_y f(a,b)(y-b) + f(a,b)$$

which looks a lot like the equation for the tangent line, except now there are two slopes and two variables that need to be taken into account.

Now, what about taking multiple partial derivatives? In principle one can take both $\partial_x \partial_y f(x, y)$ and $\partial_y \partial_x f(x, y)$. Are these the same? Are we detecting the same

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change in both x and y in both cases? In general, no we are not, but in every case that we'll run into during the GRE, yes. The reason is Clairaut's theorem:

Theorem 3.11 (Clairaut's Theorem)**.** Suppose that $f\colon \mathbb{R}^2 \to \mathbb{R}$ is a function, and suppose that the second-order partial derivatives of f exist and are continuous in a neighbourhood of (a, b). Then $\partial_x \partial_y f(a, b) = \partial_y \partial_x f(a, b)$.

This is convenient and not necessarily expected, but it does make a particular technique in optimisation a whole lot more convenient later on. This also works in more than 2 variables when taking partial derivatives with respect to any two different independent variables.

3.5. **Gradient and directional derivatives.** Having done partial derivatives, it's fair to ask if there's anything resembling a 'total derivative' of the function, something that takes all the variables into account. There is.

Definition 3.12. For $f\colon \mathbb{R}^2 \to \mathbb{R}$, define the gradient of f at (a, b) to be

$$\nabla f(a, b) = \langle \partial_x f(a, b), \partial_y f(a, b) \rangle.$$

What good is the gradient for us? It is a vector quantity instead of a scalar quantity, which is interesting, and it still satisfies some nice properties. Because it is made of partial derivatives, it is still linear, and it satisfies a product rule:

$$\nabla(f(x, y) \cdot g(x, y)) = f(x, y) \cdot \nabla g(x, y) + g(x, y) \cdot \nabla f(x, y)$$

where we think of f and g as scalar multiples (though depending on (x, y)). There is also a chain rule for the types of functions that we can actually compose at this point: suppose that $\varphi\colon \mathbb{R} \to \mathbb{R}$, so that $\varphi \circ f\colon \mathbb{R}^2 \to \mathbb{R}$ is still a multivariable function. Then

$$\nabla(\varphi \circ f) = \varphi'(f(x, y)) \cdot \nabla f(x, y)$$

where again we think of the function $\varphi'\colon \mathbb{R} \to \mathbb{R}$ as acting by scalar multiplication.

Problem 3.13. Compute the gradient of

$$g(x, y, z) = (x^2 + y^2 + z^2)^8$$

We can now talk about directional derivatives in other direction. The partial derivatives are the derivatives in the direction $\langle 1, 0 \rangle$ or $\langle 0, 1 \rangle$, but we could have used any other unit vector.

Definition 3.14. The directional derivative of $f\colon \mathbb{R}^2 \to \mathbb{R}$ at (a, b) in the direction $\vec{u} = \langle h, k \rangle$ is the limit

$$\partial_{\vec{u}} f(a, b) = \lim_{t \to 0} \frac{f(a + th, b + tk) - f(a, b)}{t}$$

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Note that these may not exist if the function is not differentiable at (a, b). In particular, the existence of partial derivatives alone is not sufficient to conclude these exist. But in the case $f(x, y)$ is differentiable, we have the following:

$$\partial_{\vec{u}} f(a, b) = \nabla f(a, b) \cdot \vec{u}$$

which means that the above limit needs to be computed only rarely.

Problem 3.15. Prove that the directional derivatives of

$$f(x, y) = \begin{cases} \frac{xy^4}{x^2+y^8} & (x, y) \neq \vec{0} \\ 0 & (x, y) = \vec{0} \end{cases}$$

at $(0, 0)$ exist and depend linearly on the gradient, but that $f(x, y)$ is not differentiable at $(0, 0)$.

Problem 3.16. Prove that there is no function $f(x, y)$ such that $\nabla f(x, y) = \langle y^2, x \rangle$. Hint: Clairaut's theorem.

We have another version of the chain rule, where we compose a curve and a multivariable function to obtain a function $\mathbb{R} \to \mathbb{R}$.

Theorem 3.17 (Chain Rule II)**.** Let $r \colon \mathbb{R} \to \mathbb{R}^2$ be a differentiable curve and let $f \colon \mathbb{R}^2 \to \mathbb{R}$ be a differentiable function. Then

$$\frac{d}{dt} f(\vec{r}(t)) = \nabla f(\vec{r}(t)) \cdot \vec{r}'(t)$$

where this is now the dot product of the two vector-valued functions.

Problem 3.18. Prove this theorem from the definition of a single variable derivative.

Gradient is in the direction of greatest change on the graph $z = f(x, y)$. How do we see this? The directional derivatives of $f(x, y)$ tell us the rate of change in each direction. The direction $\vec{u}$ which makes the quantity $\nabla f(a, b) \cdot \vec{u}$ the most is the unit vector in the direction of $\nabla f(a, b)$ itself. Similarly, $-\nabla f(a, b)$ is the direction of greatest decrease.

For another thing, suppose that we look at the level curves of the graph $z = f(x, y)$. These are the specific subsets $f(x, y) = c$ for a fixed constant $c \in \mathbb{R}$. Let $\vec{r}_c(t)$ parametrise the curve, and consider a point $(a, b) = \vec{r}_c(t_0)$ on this curve. Then the tangent vector to the curve is $\vec{r}_c'(t)$, and we can examine $\nabla f(a, b) \cdot \vec{r}_c'(t_0)$. By the chain rule,

$$\nabla f(a, b) \cdot \vec{r}_c'(t_0) = \frac{d}{dt} f(\vec{r}_c(t)) \Big|_{t=t_0}$$

But on the curve $\vec{r}_c(t)$, f is the constant c. Thus the above derivative is zero. This means that the gradient is orthogonal to the tangent vector to the level curve, i.e. it

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is normal to the level curve. This is also true in higher dimensions, though it's a bit more complicated to prove.

Problem 3.19. What is the greatest rate of change of $f(x, y) = x^4y^{-2}$ at the point $(a, b) = (2, 1)$?

The last thing to say is on the subject of surfaces on $\mathbb{R}^3$ which are defined using 3-variable functions. Consider a function $F\colon \mathbb{R}^3 \to \mathbb{R}$ and consider the set of points (x, y, z) such that $F(x, y, z) = k$. Assuming that F is a nice function, this defines a surface in $\mathbb{R}^3$, but usually one that isn't the graph of a function. If we consider the easiest example,

$$x^2 + y^2 + z^2 = 1$$

then we get a sphere, which we know isn't the graph of a function, but is the graph of two functions glued together.

3.6. **Optimisation.** Make sure to do lots of optimisation and Lagrange multipliers too, and second der test

$f(x, y) = (x^2 + y^2)e^{-x}$

max of $f(x, y) = x + y - x^2 - y^2 - xy$ on $[0, 2]^2$

Lagrange multipliers on building the perfect rectangular region

4. Day 4: Multivariable calculus

How do you integrate in two variables? First, learn how to integrate boxes. You can do that using Riemann sums, but I really don't want to do that. It's conceptually important but whatever. Again, they are linear and you can separate them up and so on.

It doesn't matter whether you integrate over x first or y.

Theorem 4.1 (Fubini's Theorem)**.** Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be continuous and let $R = [a, b] \times [c, d]$ be a rectangle. Then

$$\iint_R f(x, y)\, dA = \int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy$$

Then, for integrating functions over more complicated regions, you want to parametrise the region in terms of x in some range then y as a function of x, then integrate y first then x. Or you can do it the other way, depending on exactly how your region D looks.

Problem 4.2. Integrate $f(x, y) = xy$ over the region bounded by $y = 4$ and $y = x^2$ in the first quadrant.

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The integral is the volume under the surface in $\mathbb{R}^3$, not the area under the curve. This means that instead of calculating the area between curves, we can calculate the volume of a region between surfaces. In particular, when a certain region has a nice boundary with respect to the xy-plane and its upper and lower boundaries are nice functions of x, y, we're in business.

4.1. **Triple Integrals.** Really, there's not a whole lot different here, except that we have three variables instead of two. Riemann sums are Riemann sums, except now they're one dimension more annoying. Supposing that we can parametrise our region $W \subset \mathbb{R}^3$ as above, so that its boundary is of the form $z_1(x, y) \leq z \leq z_2(x, y)$ on a region D in the xy-plane, then

$$\iiint_W f(x, y, z)\, dV = \iint_D \left(\int_{z_1(x,y)}^{z_2(x,y)} f(x, y, z)\, dz \right) dA$$

Moreover, if we want the volume of W, we just integrate the function 1.

Problem 4.3. Evaluate $\displaystyle\iiint_W z\, dV$ where W is the region between the planes $z = x + y$ and $z = 3x + 5y$ over the rectangle $[0, 3] \times [0, 2]$.

Rogawski p883 of the pdf 863 of the book, the latter plane is on top

Problem 4.4. Integrate $f(x, y, z) = x$ over the region W bounded above by $z = 4 - x^2 - y^2$ and below by $z = x^2 + 3y^2$.

The former is on top, so need to compute the boundary of D which turns out to be an ellipse once you set them equal to each other. Not yet appropriate to use polar because we haven't learned it, but we can use the crappier version of the integral for practice.

4.2. **Change of coordinates.** Go over polar, cylindrical and spherical. It should be clear the types of regions that are suitable: polar is for round things in 2d, cylindrical is for round things that have height, and sphere is for spherical or elliptical things.

The problem is that $dr\, d\theta$ is not the same area as $dx\, dy$. In fact, we can draw the usual picture and prove that

$$dx\, dy = r\, dr\, d\theta$$

We have similar problems in cylindrical and spherical coordinates. For spherical coordinates, it's hard but possible to show that

$$dx\, dy\, dz = \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta$$

Problem 4.5. Compute the volume of the region between the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$.

In Progress

Now would probably be a good time to go over quadric surfaces:

- Ellipsoid
- Elliptic paraboloid
- Hyperbolic paraboloid
- Hyperboloid of one sheet
- Hyperboloid of two sheets

4.3. **Vector fields and fancier integration.** We now turn to the second kind of integration in multivariable calculus, namely those involving vector-valued functions.

Definition 4.6. A vector field is a function $f\colon \mathbb{R}^n \to \mathbb{R}^n$, which we think of as assigning to each point in $\mathbb{R}^n$ a vector in $\mathbb{R}^n$ beginning at that point.

At this point, I would draw a picture.

There are many vector fields in real life, two easily coming to mind are the gravitational vector field which expresses the force (and direction) due to gravity on any object in space, and on each we can talk about the vector field of wind – to each point on each we can assign the vector of which direction (and speed) the wind is blowing.

The most boring example is when we are still working with real-valued functions. If we want to integrate a real-valued function over some curve C in $\mathbb{R}^3$ (or a surface, but let's stick with curve), then we think of C as being the image of some $\gamma\colon [a,b] \to \mathbb{R}^3$ which is continuous except at finitely many points (with technical details omitted). Then the function we are considering is

$$f(\gamma(t))\colon \mathbb{R} \to \mathbb{R}$$

But this isn't quite enough, because the parametrisation matters. We need to make sure that the speed at which we are traversing this curve is taken into account, i.e.

$$\int_C f(x,y,z) = \int_a^b f(\gamma(t))\|\gamma'(t)\|\,dt$$

Note that for this equation to be without problems, we want to assume that $\gamma'(t)$ and $f(x,y,z)$ are continuous. But this is the boring example.

Now, suppose we are thinking physics, and we want to know something like 'how much energy does it take to fight gravity or the wind'? This would involve a vector field $F\colon \mathbb{R}^3 \to \mathbb{R}^3$. In this case, whenever the curve travels in the same direction as the vector field, we would like to value that positively (going with the flow), and negative when the curve travels against it. Remember that C we cannot think of as just a 1-dimensional object in $\mathbb{R}^3$, it comes with an orientation – it has a back and a front, and the function $\gamma\colon [a,b] \to \mathbb{R}^3$ we use needs to take this into account.

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What vector operation determines whether things go in the same direction? The dot product. What gives the (linear) direction the curve is going? Its tangent vectors $\gamma'(t)$. Thus:

Definition 4.7. The line integral of a vector field F along a curve C, parametrised by $\gamma\colon [a,b] \to \mathbb{R}^3$, is

$$\int_C F \cdot dr = \int_a^b F(\gamma(t)) \cdot \gamma'(t)\, dt$$

Both of these quantities are vectors, so it makes sense to dot them. Another way the expression $F \cdot dr$ is sometimes written is $F_1\, dx + F_2\, dy + F_3\, dz$, where these are the component functions of F. This will come up later.

Problem 4.8. Compute the line integral of $F = \langle z, y^2, x\rangle$ along the curve $\gamma(t) = (t+1, e^t, t^2)$ for $t \in [0,2]$.

What are some basic properties of the line integral? They are still linear, and now if one reverses the orientation of the curve C, this is like swapping a, b on the righthand side of the above equation, hence negates the integral. The last thing is that stringing together multiple curves end-to-end gives a sum of integrals.

4.4. **Conservative vector fields.** These are just the best. Our prototype here is a vector field F that arises as ∇f for some $f\colon \mathbb{R}^3 \to \mathbb{R}$. Such a vector field is called conservative. Then we can use the fundamental theorem of calculus to evaluate

$$\int_C F \cdot dr = \int_a^b \nabla f(\gamma(t)) \cdot \gamma'(t)\, dt = \int_a^b \frac{d}{dt} f(\gamma(t))\, dt = f(\gamma(b)) - f(\gamma(a))$$

But $\gamma(b)$ and $\gamma(a)$ are just the endpoints of the curve, so the actual curve C doesn't matter in this case. Any vector field for which this happens is called path-independent.

Similarly, if C is a closed curve, its endpoints are the same, so

$$\oint_C F \cdot dr = 0$$

Problem 4.9. Verify that $F(x,y,z) = \langle 2xy + z, x^2, x\rangle$ is the gradient of a function, then evaluate the line integral over the curve $\gamma(t) = \langle \sin(t)\cos(\pi t), e^t, 4t^3 - 1\rangle$ for $t \in [0, 1/2]$.

One might ask if there are path-independent vector fields that do not arise as the gradient of some function. The answer is, essentially, no.

Theorem 4.10. A vector field F on an open, connected domain D is path-independent if and only if it is conservative.

In Progress

Now, let us go over some of the other vector derivatives that will become useful shortly. The first is the divergence of a vector field,

$$\operatorname{div} F(x,y,z) = \nabla \cdot F = \partial_x F_1 + \partial_y F_2 + \partial_z F_3$$

and the second is the curl of a vector field, which only makes sense in $\mathbb{R}^3$:

$$\operatorname{curl} F(x,y,z) = \nabla \times F = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{bmatrix}$$

Problem 4.11. Prove that if $F = \nabla f$ is conservative, then $\operatorname{curl} F = \vec{0}$.

The converse to this problem is not always true, but it is true in a great many cases.

Theorem 4.12. Suppose that D is an open simply-connected domain. A vector field F on D is conservative if and only if $\operatorname{curl} F = \vec{0}$.

Hence one should beware that the domains they are working on be simply-connected, which (we remember) means that all loops in D be contracted to a point.

4.5. **Surface integrals.** In order to integrate using surfaces in $\mathbb{R}^3$, we need to be able to parametrise them like

$$G(u,v) = (x(u,v), y(u,v), z(u,v))$$

where (u,v) are in some region D in $\mathbb{R}^2$. For example, the graph of the function $z = f(x,y)$ in $\mathbb{R}^3$ is easily parametrised by $(x, y, f(x,y))$. This is our prototype, but of course not all surfaces in $\mathbb{R}^3$ are graphs.

Suppose we want to parametrise the cylinder $x^2 + y^2 = 1$ in $\mathbb{R}^3$. This is best done with cylindrical coordinates (of course), yielding an easy parametrisation $(1, \theta, z)$ for $\theta \in [0, 2\pi]$ and $z \in \mathbb{R}$. Since the radius is fixed, we only get two variables.

We can parametrise spheres similarly using spherical coordinates. There's only a slight problem with this picture, as we get kind of an overlap at $\theta = 0$ and $\theta = 2\pi$, but we will not concern ourselves overmuch with this.

Okay, what are we doing with surfaces? Given a vector field, we want to measure how much the vector field is flowing through the surface. But what does flowing through the surface mean? Do surfaces have an up side and a down side? It turns out, they do. In the parametrisation here, we have two tangent vectors $\partial_u G$ and $\partial_v G$ which naturally give a direction 'up' for the surface in the form of $\partial_u G \times \partial_v G$. Note that if the partial derivatives are parallel this breaks, so we want to make sure that we don't run into that problem.

Once we have that normal vector, we can begin by finding the tangent plane, which is moderately useful.

In Progress

Problem 4.13. Compute the tangent plane to the surface $G(\theta, z) = (2\cos\theta, 2\sin\theta, z)$ at $P = G(\pi/4, 5)$.

Of course, we could easily reverse the orientation on the parametrisation by using the equation $G(z, \theta)$ so that the order in which we take the cross product is swapped. Hence we're going to want to have some sort of consistency. For the surface given by the graph of a function, we will take the upwards direction to the canonical orientation for the normal vector, i.e. the one going in the positive z-direction $\langle -\partial_x, -\partial_y, 1\rangle$.

Now, how do we perform scalar surface integrals, i.e. ones that ignore the vector field for the moment? Turns out that, in the above situation, $N = \partial_u G \times \partial_v G$ is also capturing the infinitesimal area of the parallelogram with sides du, dv. This is an easy computation using the $\sin\theta$ interpretation of the cross product. Therefore if we want to do our integral, we need to scale by this amount, just like we had to account for the speed in the case of line integrals.

Definition 4.14. Let $G(u, v)$ be a parametrisation of a surface $S \subset \mathbb{R}^3$ with domain D. Assume that G is C^1, one-to-one, and regular (i.e. the normal vector is nondegenerate). Then for a function $f\colon \mathbb{R}^3 \to \mathbb{R}$,

$$\iint_S f(x, y, z)\, dS = \iint f(G(u, v))\|\partial_u G(u, v) \times \partial_v G(u, v)\|\, du\, dv$$

Now, once we bring in the actual flow and a vector field $F\colon \mathbb{R}^3 \to \mathbb{R}^3$, we need to use a dot product:

Theorem 4.15.

$$\iint_S F \cdot dS = \iint_D F(G(u, v)) \cdot N(u, v)\, du\, dv$$

where again $N(u, v) = \partial_u G(u, v) \times \partial_v G(u, v)$.

Note that we usually want to make sure that $N(u, v)$ is the upward-pointing (or outward-pointing in the case of a closed surface) normal vector, just like in the graph case. Therefore one might need to shuffle coordinates around if we accidentally have gotten it wrong.

Problem 4.16. Compute the flux through the surface $G(u, v) = (u^2, v, u^3 - v^2)$ over $D = [0, 1]^2$ of the vector field $F = \langle 0, 0, x\rangle$.

P981 Rog, book page 961

4.6. **Fundamental theorems.** Green's Theorem

Theorem 4.17 (Green's Theorem)**.** Let D be a closed domain with ∂D a simple closed curve, oriented counterclockwise. Then

$$\oint_{\partial D} F_1 dx + F_2 dy = \iint_D (\partial_x F_2 - \partial_y F_1)\, dA$$

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Use this when the line integral of the closed curve would be way too confusing to compute.

Problem 4.18. Verify Green's Theorem by computing the line integral over the unit circle C of $F(x, y) = (xy^2, x)$.

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One key feature is that, even if your curve is not closed, you can close it up and appeal to a simpler area calculation, i.e. if you have to compute a line integral over half a circle, complete it to a whole circle, etc.

Stokes' Theorem

Theorem 4.19 (Stokes' Theorem)**.** Let S be an oriented surface with boundary oriented so that the surface is always on your left (assuming outward pointing normal vectors). Assume that F is a C^1 vector field. Then

$$\oint_{\partial S} F \cdot dr = \iint_S \operatorname{curl} F \cdot dS$$

In particular, if $\partial S = \varnothing$, then the integral is zero.

Also add the thing about surfaces having different boundaries, and $\div \operatorname{curl} = 0$ $\div = 0$ if and only if it's the curl of something in a simply connected domain.

Divergence Theorem

5. Day 5: Linear algebra

Right, systems of linear equations. We like solving them, don't we? Let's just cut to the chase. We already know well enough what a vector space is and what an inner product and norm are.

Do subspaces and stuff too

Now, though, we want to open up inner products a bit. Over $\mathbb{C}$, we want to define the inner product $\langle \vec{v}, \vec{w} \rangle$ to still be the sum $\sum_{i=1}^n v_i w_i$, but now we want it to be sesquilinear:

$$\langle \alpha \cdot \vec{v}, \vec{w} \rangle = \langle \vec{v}, \overline{\alpha} \cdot \vec{w} \rangle, \quad \alpha \in \mathbb{C}$$

Inner products and norms satisfy what's called the Cauchy-Schwartz inequality: for any $\vec{v}, \vec{w} \in V$,

$$|\langle \vec{v}, \vec{w} \rangle| \leq \|\vec{v}\| \cdot \|\vec{w}\|$$

and equality is only satisfied in certain cases. We also have the triangle inequality:

$$\|\vec{v} + \vec{w}\| \leq \|\vec{v}\| + \|\vec{w}\|$$

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Both of these can be proven using the idea of projections. We define the projection of $\vec{w}$ onto $\vec{v}$ by

$$\operatorname{proj}_{\vec{w}} \vec{v} = \frac{\langle \vec{v}, \vec{w} \rangle}{\langle \vec{w}, \vec{w} \rangle} \cdot \vec{w}.$$

That's a good formula to remember, and does things like proves the $\cos\theta$ formula for inner product.

One fun formula that you might recall is the polarization identity. In any real vector space,

$$\langle \vec{v}, \vec{w} \rangle = \frac{1}{4} \left(\|\vec{v} + \vec{w}\|^2 - \|\vec{v} - \vec{w}\|^2 \right)$$

which is an easy proof. Make it a PROBLEM or just do it, reminding us that we should think of norms as inner products squared whenever possible.

A set in a vector space is linearly independent if $\sum a_i \vec{v_i} = 0 \implies a_i = 0$ for all i. A maximal linearly independent set is called a basis. All bases have the same number of elements, and all (finite dimensional) vector spaces have a basis. Note that infinite dimensional vector spaces don't have a basis unless you assume the axiom of choice!

What's the best kind of basis? An orthonormal one, that is, the basis elements are pairwise orthogonal (inner product equals zero) and have length 1. Luckily, any (finite) basis can be processed into an orthonormal one using the Gram-Schmidt process. THEN EXPLAIN THE PROCESS

Problem 5.1. Let $\{1, x, x^2, x^3\}$ be a basis for $P_3(\mathbb{R})$. Convert it to an orthonormal basis, where we let $\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x)\,dx$.

What are the functions we care about?

Definition 5.2. A linear transformation $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is a function satisfying the axiom that $T(c \cdot \vec{v} + \vec{w}) = c \cdot T(\vec{v}) + T(\vec{w})$ for all $c \in \mathbb{R}$ and for all $\vec{v}, \vec{w} \in \mathbb{R}^n$.

The definition implies that $T(\vec{0}) = \vec{0}$, which is a nice feature. Linear transformations are well described by matrices: T represents a matrix $A \in M_{m\times n}(\mathbb{R})$ where the columns of A are $T(\vec{e_i})$ for the basis vectors $\vec{e_i}$ of $\mathbb{R}^n$. The rank of T is the dimension of its image, and the nullity the dimension of its kernel.

Theorem 5.3. Rank + nullity $= n$, the dimension of the domain.

Similar matrices conjugation by an invertible. How do we change basis? Consider the basis $\beta = \{\vec{b_i}\}$. Then in order to write the matrix A in terms of the basis β, we think of converting from standard to β, doing the transformation, then back to standard. The 'back to standard' matrix is P such that its columns are the $\vec{b_i}$. Therefore

$$A = P^{-1}BP \implies B = PAP^{-1}$$

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where the matrix P is easy to compute but P^{-1} we have to use Gaussian elimination (see below).

What does it mean to be invertible? First, need to be $n \times n$, so that there's a chance that both $AB = I_n$ and $BA = I_n$. You can't have it both ways! That's rank-nullity theorem.

Over $\mathbb{R}$, it means that $\det \neq 0$. What's determinant? Well, go over expansion by minors equation. There's some intrinsic definition using fancier math but never mind that.

Equivalent formulations of invertibility:

- $\det A \neq 0$
- row rank of A is n
- column rank of A is n
- the linear transformation that A defines is bijective
- reduced row echelon form is the identity

How do you find the inverse of a matrix? In 2×2, there's a nice formula, but otherwise you have to use Gaussian elimination. THEN DO A 3×3 EXAMPLE.

Define eigenvalues and eigenvectors of a matrix. Note that the kernel is eigenvectors of eigenvalue 0. Sometimes you get a basis of eigenvectors, but sometimes you don't. How often does that occur?

Theorem 5.4 (Spectral Theorem)**.** This is a restricted version, but often good enough. If A is a real matrix and $A = A^{\mathrm{T}}$, the transpose of A, then A is diagonalizable. If A is a complex matrix, then we need $A = A^*$, the Hermitian matrix (transpose conjugate). The general form is that $AA^{\mathrm{T}} = A^{\mathrm{T}}A$.

So how do we go about finding eigenvalues or eigenvectors? Use the characteristic polynomial or, as came up on the test, just examine the matrix $A - \lambda I_n$. λ is an eigenvalue if and only if that matrix is not invertible, i.e. has a kernel so that $A\vec{v} - \lambda \cdot \vec{v} = 0$ has a solution and $A\vec{v} = \lambda \cdot \vec{v}$.

DO AN EXAMPLE OF FINDING EIGENVALUES FOR A SYMMETRIC 3×3 matrix.

Note: A and A^{T} have the same eigenvalues, which is pretty neat.

RANDOM SHIT: if A is nilpotent, then $I_n + A$ or $I_n - A$ is invertible (by the formula that appeared before).

Simultaneously diagonalizable: suppose that A, B have the same eigenbasis. Then $AB = BA$.

6. Day 6: Differential equations and complex analysis

So we begin today by recalling the basic types of differential equations that we need to solve for the GRE. There aren't that many, but if you're like me, you've definitely

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forgotten about them. But before we get there, let's recall the fundamental theorem that lets us do anything:

Theorem 6.1. Suppose that $f(t, y)$ and $\partial_y f(t, y)$ are continuous on a compact subset $K \subset \mathbb{R}^2$. Then for any point $(t_0, y_0) \in K$, the differential equation $y' = f(t, y)$, $y(t_0) = y_0$ has a unique solution in some neighbourhood of (t_0, y_0).

So perhaps it's good to check that $\partial_y f(t, y)$ is continuous before blindly charging into a problem, but probably not.

The first type of differential equation is the basic separation of variables, like so:

Problem 6.2. Suppose that a colony of bacteria grows at a rate directly proportional to its size. Initially, the colony has 100 bacteria, and after a week it has 300 bacteria. Write a formula modelling this situation where the time t is measured in days

Solution. The situation we have is

$$\frac{dB}{dt} = k \cdot B \implies \frac{dB}{B} = k\,dt$$

Then integrating both sides gives us

$$\log(B) = k \cdot t + C \implies B(t) = C \cdot e^{kt}.$$

We have that $B(0) = 100$, so $C = 100$. We also know that $B(7) = 300$, so

$$e^{7k} = 3 \implies k = \frac{\log(3)}{7}.$$

Putting it all together,

$$B(t) = 100e^{\frac{\log(3)}{7} \cdot t}$$

There are more sophisticated version of this problem, and they are usually in the tune of salty or sugary tanks of water.

Problem 6.3. Suppose that we have a 100L tank with 50L of water and 100g of salt in it. Suppose the tank drains at a rate of 1L/m and is filled at a rate of 2L/m with pure water. Assuming instantaneous mixing, when the tank is full, how much salt is there in the tank?

Solution. Let's set this up. We have

$$\frac{dS}{dt} = \text{in} - \text{out}$$

In this example, there's no salt coming in. For the out, we need to know what the density of salt in the tank is. The amount of salt is S, but the volume changes: we have a net +1L/m, so the volume is $50 + t$. Therefore

$$\frac{dS}{dt} = -\frac{S}{50+t} \implies \frac{dS}{S} = \frac{-dt}{50+t} \implies \log(S) = -\log(50+t) + C$$

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After integration and rearranging (specifically after pulling in the -1 into the log), we get

$$S(t) = \frac{C}{t+50}$$

The amount of salt at the beginning is 100, so we have $S(0) = C50 = 100$ so $C = 5000$. The tank is full at $t = 50$, so

$$S(t) = \frac{5000}{t+50} \implies S(50) = \frac{5000}{100} = 50\text{g}$$

Now, we turn to other types of differential equations. Let's first recall what an integrating factor is. Suppose our differential equation is of the form

$$\frac{dy}{dt} + p(t)y = q(t).$$

Then we consider the integrating factor $\mu(t) = e^{\int p(t)\,dt}$. Why does that help? Using this term,

$$\mu(t) \cdot y' + \mu(t)p(t) \cdot y = \frac{d}{dt}(\mu(t) \cdot y) = \mu(t) \cdot q(t).$$

Thus, when we integrate both sides,

$$\mu(t) \cdot y = \int \mu(t)q(t)\,dt$$

Assuming that the righthand side is integrable, we can then solve and divide out by $\mu(t)$.

Problem 6.4. Solve the linear ODE $y' - 2ty = t$.

Solution. The process implies that $\mu(t) = e^{\int -2t\,dt} = e^{-t^2}$, not something we can integrate on its own. Luckily, the whole righthand side is integrable:

$$\int te^{-t^2}\,dt = -\frac{1}{2}e^{-t^2} + C$$

Dividing through now by our integrating factor,

$$y(t) = Ce^{t^2} - \frac{1}{2}$$

Again, if we get a linear ODE of this form, this is pretty much the only way to solve it. Exact ODEs likely won't come up, but there's always that chance. Plus, it's related to multivariable calculus. Suppose that we have a differential equation of the form

$$N(x,y) \cdot y' + M(x,y) = 0 \text{ i.e. } N(x,y)dy + M(x,y)dx = 0$$

where moreover we have $\partial_x N(X,y) = \partial_y M(x,y)$. Then this implies that, at least locally, that this situation is coming from the equality is mixed partials, so we need to

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find a function $H(x, y)$ with $\nabla H = \langle M, N \rangle$. The general solution to the differential equation is $H(x, y) = C$.

Problem 6.5. Solve $(x^2y + 2y) \cdot y' + (xy^2 + 2x) = 0$.

Solution. This equation is exact (easily verified), so a solution looks something like

$$H(x, y) = \int xy^2 + 2x \, dx = \int x^2y + 2y \, dy$$

As before, we need to integrate but bear in mind that we might have constants that depend on one variable or the other. That is,

$$\int xy^2 + 2x \, dx = \frac{x^2y^2}{2} + x^2 + g_1(y), \quad \int x^2y + 2y \, dy = \frac{x^2y^2}{2} + y^2 + g_2(x)$$

Comparing terms, we need to use $g_2(x) = x^2$ and $g_1(y) = y^2$, so that

$$H(x, y) = \frac{x^2y^2}{2} + x^2 + y^2 = C$$

is our general solution.

6.1. **Higher order differential equations.** Now, suppose we want to solve particular differential higher order differential equations that have little interaction between the variables. For instance, examine

$$y'' - 9y = f(t)$$

The first step is to solve the corresponding homogeneous equation $y'' - 9y = 0$. We can solve this by inspection, know that $y' = ky$ is solved by e^{kt}. Hence the solutions we need are e^{3t} and e^{-3t}. The general solution to the differential equation is therefore

$$y(t) = c_1e^{3t} + c_2e^{-3t}.$$

Here's generally how you solve a homogeneous differential equation like this. Consider an equation

$$ay'' + by' + cy = 0$$

Then solutions to this equation are given by $e^{\lambda t}$, where λ is a root of the corresponding characteristic polynomial

$$ax^2 + bx + c = 0$$

There are three options here: the polynomial may have two distinct real roots, one double real root, or two (conjugate) complex roots.

The case of two distinct real roots is the one we examined above: the general solution is $y(t) = c_1e^{\lambda_1 t} + c_2e^{\lambda_2 t}$. If there is only one real root, we still need a two-dimensional solution to the system of equations, so the general solution looks like $y(t) = c_1e^{\lambda t} + c_2te^{\lambda t}$. The complex roots possibility is a little more delicate, because we need to make sure that come up with a real solution.

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To examine this, let $\lambda = a + bi$. Then the general solution becomes

$$y(t) = c_1 e^{(a+bi)t} + c_2 e^{(a-bi)t} = c_1 e^{at} e^{i \cdot bt} + c_2 e^{at} e^{i \cdot (-bt)}$$

Using the identity $e^{i\theta} = \cos\theta + i \sin\theta$, we change the above:

$$\begin{aligned} y(t) &= c_1 e^{at} e^{i \cdot bt} + c_2 e^{at} e^{i \cdot (-bt)} \\ &= c_1 e^{at}(\cos(bt) + i \sin(bt)) + c_2 e^{at}(\cos(-bt) + i \sin(-bt)) \end{aligned}$$

Using now that $\cos(-bt) = \cos(bt)$ and $\sin(-bt) = -\sin(bt)$,

$$y(t) = (c_1 + c_2) e^{at} \cos(bt) + (c_1 - c_2) i \cdot e^{at} \sin(bt)$$

Now we use the fact that (secretly) $c_1, c_2 \in \mathbb{C}$ now, we need that $c_1 - c_2 \in i\mathbb{R}$ and $c_1 + c_2 \in \mathbb{R}$. Luckily, it is possible to get any number we want using $c_1 = \frac{c-di}{2}$ and $c_2 = \frac{c+di}{2}$ so that $c_1 + c_2 = c$ and $c_1 - c_2 = -di$. Putting this all together, the general solution is

$$y(t) = c_1 e^{at} \cos(bt) + c_2 e^{at} \sin(bt).$$

Problem 6.6. Solve the following initial value problem: $y'' - 4y' + 9y = 0$, with $y(0) = 0$ and $y'(0) = -8$.

Solution. The characteristic equation is $x^2 - 4x + 9 = 0$, so that we have roots

$$\lambda = \frac{4 \pm \sqrt{16 - 4(9)}}{2} = 2 \pm \sqrt{5} i$$

Hence the general solution is

$$y(t) = c_1 e^{2t} \cos(\sqrt{5} t) + c_2 e^{2t} \sin(\sqrt{5} t)$$

Knowing that $y(0) = 0$ means that $c_1 = 0$, as everything else cancels out. Therefore

$$y(t) = c e^{2t} \sin(\sqrt{5} t) \text{ and } y'(t) = 2ce^2 \sin(\sqrt{5} t) + \sqrt{5} c e^{2t} \cos(\sqrt{5} t)$$

So $y'(0) = \sqrt{5} \cdot c = -8$ thus $c = -\dfrac{8}{\sqrt{5}}$. Not the nicest solution, but

$$y(t) = -\frac{8}{\sqrt{5}} e^{2t} \sin(\sqrt{5} t)$$

6.2. **Nonhomogeneous differential equations.** How do we deal with nonhomogeneous differential equations? First, solve the homogeneous one. Then we have to guess a particular solution.

Problem 6.7. Determine a particular solution to $y'' - 4y' - 12y = 3e^{5t}$.

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Solution. We need to solve $x^2 - 4x - 12 = 0$, which isn't too difficult: it factors as $(x-2)(x+6) = 0$ so we get

$$y(t) = c_1 e^{-6t} + c_2 e^{2t} + y_p(t)$$

What should $y_p(t)$ look like? Probably something of the form $y_p(t) = Ae^{5t}$, so we then need to check which A satisfies the differential equation:

$$y_p'(t) = 5Ae^{5t},\ y_p''(t) = 25Ae^{5t} \implies 25Ae^{5t} - 4 \cdot 5Ae^{5t} - 12 \cdot Ae^{5t} = 3e^{5t}$$

Solving this gives $-7Ae^{5t} = 3e^{5t}$ so $A = -\dfrac{3}{7}$. Putting this all together,

$$y(t) = c_1 e^{-6t} + c_2 e^{2t} - \frac{3}{7} e^{5t}$$

Other types of particular solutions require different guesses: sines and cosines demand sines and cosines. What if particular solutions are polynomials?

Problem 6.8. Determine a particular solution to $y'' - 4y' - 12y = t^2 + 3t + 2$.

Solution. The particular solution looks like a polynomial of the same degree, so let $y_p(t) = at^2 + bt + c$. Then

$$y_p'(t) = 2at + b, \quad y_p''(t) = 2a$$

Putting it all together,

$$2a - 4(2at + b) - 12(at^2 + bt + c) = t^2 + 3t + 2$$

We need to separate by degrees:

$$-12at^2 = t^2, \quad (-8a - 12b)t = 3t, \quad 2a - 4b - 12c = 2$$

The easiest way to solve this is left to right:

$$a = -\frac{1}{12}, \quad \left(\frac{8}{12} - 12b\right) = 3 \implies 12b = -\frac{7}{3} \implies b = -\frac{7}{36}$$

Finally, we can solve that $c = -\dfrac{25}{216}$. We can then put it all together as we did above.

Fortunately, this is as far as things need to go in the realm of differential equations.

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6.3. **Complex analysis.** Let's recall a little the nice types of complex-valued functions.

Theorem 6.9. Let $f\colon \Omega \to \mathbb{C}$, where $\Omega \subset \mathbb{C}$ is an open subset of the complex numbers. Then the following are equivalent:

- f is differentiable in an open disc centered at $a \in \Omega$ (holomorphic)
- f has a convergent power series expansion $\sum_{n=0}^{\infty} c_n(z-a)^n$ in an open disc centered at $a \in \Omega$ (analytic)

This incredible theorem implies that differentiable functions are smooth, which is one of our introductions to the wild world of complex analysis. There are some nice corollaries:

Corollary 6.10. Let $f, g\colon \Omega \to \mathbb{C}$ be two holomorphic functions on an open connected $\Omega \subset \mathbb{C}$. If $f(z) = g(z)$ on an infinite subset $S \subset \Omega$ that contains a limit point of Ω, then $f = g$ on Ω.

Corollary 6.11. A bounded holomorphic function $f\colon \mathbb{C} \to \mathbb{C}$ must be constant.

Holomorphic functions must satisfy the Cauchy-Riemann equations, and the converse is true as well.

Theorem 6.12. Let $f\colon \Omega \to \mathbb{C}$ be a function, and write $f(x+iy) = u(x+iy) + i \cdot v(x+iy)$. Then f is holomorphic if and only if

$$\partial_x u = \partial_y v \text{ and } \partial_y u = -\partial_x v.$$

This theorem is pretty useful, as it means that information about the real part of a holomorphic function can get us the whole function.

Problem 6.13. Suppose that $f(x+iy) = u(x,y) + i \cdot v(x,y)$. If $u(x,y) = x^2 - y^2$ and $v(1,1) = 2$, find $v(4,1)$.

Solution. We can use the fundamental theorem of calculus in this case:

$$v(4,1) - v(1,1) = \int_1^4 \partial_x v(x,y)\, dx$$

The equation is, what's $\partial_x v$? By the Cauchy-Riemann equations, it's $-\partial_y u = 2y$. Thus

$$v(4,1) - v(1,1) = \int_1^4 2y\, dx = 2xy\Big|_{(1,1)}^{(4,1)} = 6$$

This implies that $v(4,1) - 2 = 8$ so $v(4,1) = 8$.

It might be worth remembering the following:

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Definition 6.14. A function $u(x, y)\colon \Omega \to \mathbb{R}$ is harmonic if $\partial_x^2 u + \partial_y^2 u = 0$.

Locally, a harmonic function is the real part of a holomorphic function, in the following way: if $u(x, y)$ is harmonic, then

$$f(x + iy) = \partial_x u(x, y) - i \cdot \partial_y u(x, y)$$

is holomorphic.

6.4. **Cauchy integral formula.** The last thing we should recall is the Cauchy integral formula.

Theorem 6.15 (Cauchy integral formula)**.** Suppose that $f\colon \Omega \to \mathbb{C}$ is a holomorphic function on an open domain, and let $D \subset \Omega$ be a closed disc in Ω. Then

$$f(a) = \frac{1}{2\pi i} \oint_{\partial D} \frac{f(z)}{z - a}\, dz$$

for every $a \in D$.

This yields the residue theorem.

Theorem 6.16 (Residue theorem)**.** Let $U \subset \mathbb{C}$ be a simply connected open subset and $f\colon U \to \mathbb{C}$ a function holomorphic but for $a \in U$. Let γ be a closed curve in U around a, oriented counterclockwise. Then

$$\oint_\gamma f(z)\, dz = 2\pi i \operatorname{Res}(f, a)$$

where $\operatorname{Res}(f, a)$ is the coefficient of the term $\dfrac{1}{z - a}$ in the Laurent series expansion of $f(z)$ around a. Otherwise put, it is the number R such that

$$f(z) - \frac{R}{z - a}$$

has an analytic antiderivative in a disc around a.

I'm not sure that this has much of a place on the GRE, but the Cauchy-Riemann equations are the jewel in the crown.

7. Day 7: Algebra

Topics covered: groups, rings, and fields.

In Progress

7.1. **Groups.**

Definition 7.1. A group is a set G with a binary operation $\cdot\colon G \times G \to G$ that satisfies the following axioms:

- The operation is associative, so $(g \cdot h) \cdot k = g \cdot (h \cdot k)$ for any $g, h, k \in G$
- There exists an element $e \in G$ such that $e \cdot g = g \cdot e = g$ for all $g \in G$
- For every element $g \in G$, there exists an element $g^{-1} \in G$ so that $g \cdot g^{-1} = g^{-1} \cdot g = e$

Problem 7.2. Prove that the identity element is unique.

Problem 7.3. Prove that the inverse of an element g is unique.

These are two good exercises to get your hands on. Note that the group operation needn't be commutative! Consider an example you already know: let $\mathrm{GL}_n(F)$ be the subset of invertible $n \times n$ matrices with entries in a field F. Then this is a group under multiplication, with inverse and identity as one would imagine. Note that $M_n(F)$ is *not* a group under multiplication, as non-invertible matrices do not have inverses (obviously). However, $M_n(F)$ is a group under addition, and it's in fact commutative, i.e. $A + B = B + A$ for all $A, B \in M_n(F)$. Commutative groups are also called abelian.

What are the kinds of functions we're interested in?

Definition 7.4. A group homomorphism is a map of sets $\varphi\colon G \to H$ such that $\varphi(g_1 \cdot g_2) = \varphi(g_1) \cdot \varphi(g_2)$ for all $g_1, g_2 \in G$.

Problem 7.5. Prove that $\varphi(g^{-1}) = \varphi(g)^{-1}$ and $\varphi(e_G) = e_H$.

Definition 7.6. An isomorphism of groups is a group homomorphism that is bijective as a map of sets. In particular, the set-theoretic inverse map is automatically a group homomorphism.

Problem 7.7. Prove it.

Definition 7.8. A subset $H \subset G$ is called a subgroup of G if $e_g \in H$ and for every $h_1, h_2 \in H$, $h_1 h_2^{-1} \in H$. That is, H includes the identity element and is closed under multiplication and inverses. We usually write $H < G$ in this case.

As an example of subgroups, let $g \in G$ and consider the set $\{g^n : n \in \mathbb{Z}\}$. Then it's an easy check that this satisfies the subgroup definition, and we write $\langle g \rangle < G$. Such a subgroup is called cyclic. Note that this set may be finite. If it is, we write $|g| = n$ for the order of g, and it's the minimal n such that $g^n = e_G$. Otherwise we say $|g| = \infty$.

There's a really important theorem on the order of subgroups (and hence of elements).

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Theorem 7.9 (Lagrange's Theorem)**.** Let G be a finite group and let $H < G$ be a subgroup. Then $|H|$ divides $|G|$. In particular, $|g|$ divides $|G|$ for every $g \in G$.

Hence when we see GRE questions about the possible orders of elements and subgroups, this helps a lot.

We can talk about the subgroup generated by a number of elements $g_1, \ldots, g_m$ in the obvious way. It's helpful to use the fact that the intersection of subgroups is still a subgroup, and we can define

$$\langle g_1, \ldots, g_m \rangle = \bigcap_{g_1, \ldots, g_m \in H} H.$$

Similarly, we can talk about the subgroup generated by a family of subgroups,

$$HK = \{hk : h \in H, k \in K\} = \bigcap_{H, K \subset G'} G'$$

Since $H < HK$ and $K < HK$, this means that $|HK|$ needs to be a divisor of $|H| \cdot |K|$. In particular, $|HK| = |H| \cdot |K|$ if and only if $H \cap K = \{e_G\}$.

Problem 7.10. Prove it.

Definition 7.11. A normal subgroup is a subgroup $N < G$ such that $gNg^{-1} \subset N$ for all $g \in G$. In this case, we write $N \triangleleft G$.

Normal subgroups are very important. In particular, kernels of group homomorphisms are normal subgroups. Additionally, these are the appropriate objects in order to define quotients. For any $H < G$, we define

$$G/H = \{gH : g \in G\}$$

where $g_1 H = g_2 H$ if these subsets contain the same elements, i.e. there exists $h \in H$ such that $g_1 \cdot h = g_2$. In the case that $N \triangleleft$ is a normal subgroup, G/N actually admits a group structure – $g_1 N \cdot g_2 N = g_1 g_2 N$.

Problem 7.12. Prove it.

Definition 7.13. A group G is called simple if it has no normal subgroups besides $\{e\}$ and G.

There are a variety of simple groups, but the biggest class of examples is C_p for the primes p. Another choice will turn out to be A_n for $n \geq 5$, which we will define below.

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7.2. **Examples of groups.** It's probably about time to give some examples of (finite) groups. For every positive integer $n \in \mathbb{N}$, consider the set with n elements $X_n = \{1, \ldots, n\}$. Consider a bijection $f\colon X_n \to X_n$. We can put a group structure on this set, with the operation being composition. Identity and inverses are obvious. Call the set of these maps S_n and call it the symmetric group on n elements. Then $|S_n| = n!$, one can readily check.

We think about elements in S_n using a cycle decomposition. Let $n = 5$ for simplicity, and consider the following function:

$$f(1) = 2, \quad f(2) = 3, \quad f(3) = 5, \quad f(4) = 1, \quad f(5) = 4$$

We write this in the following format: we start by writing $(1-)$, and we then write the image of 1 to obtain $(12-)$, and so on until we get (12354). This is called a 5-cycle as it's written with 5 elements. Consider another function,

$$g(1) = 2, \quad g(2) = 3, \quad g(3) = 1, \quad f(4) = 5, \quad f(5) = 4$$

which yields up the cycle decomposition $(123)(45)$, which we call a 3-2-cycle. As a final example, consider

$$h(1) = 2, \quad h(2) = 1, \quad h(3) = 3, \quad h(4) = 4, \quad h(5) = 5$$

We could write this as $(12)(3)(4)(5)$, but we'd rather write (12) and call it a 2-cycle. Note that in a cycle decomposition, the elements in the cycles must be disjoint. Every element of S_n has such a unique cycle decomposition up to permutation, and it's unique if we orient the cycle to begin with the lowest number left. That is

$$(123)(45) = (231)(54) = (312)(45)$$

but the first choice is canonical.

How do we multiply cycles? Consider $(12)(13)$. This is a composition that says $1 \to 2$, $2 \to 1 \to 3$, and $3\ to 1$. This is the cycle (123). Consider now $(13)(12)$. This says $1 \to 3$, $3 \to 1 \to 2$, and $2 \to 1$. Hence this is the cycle (132). These are different! The symmetric group S_n is not commutative. Now, we can address subgroups and orders.

Problem 7.14. The order of an m_1-m_2-$\cdots$-m_k-cycle is $\operatorname{lcm}(m_1, m_2, \ldots, m_k)$.

As such, there isn't an obvious formula for the maximal order of a cycle in S_n, but is easily computed for a given n.

There are some interesting subgroups of S_n that we can get using both group theory and geometry. The first geometric subgroup is the cyclic group C_n, which is generated by any n-cycle in S_n. All such groups are symmetric. This represents rigids rotations of the regular n-gon. Of course, there's another symmetry of the regular n-gon, which is the flip along a vertical axis of symmetry. It's harder to write down a cycle decomposition, but we can do it for $n = 5$. Let the cycle be

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$(12345) = \sigma$, and then the flip is given by $(25)(34) = \tau$. The group generated by these two elements has $2n$ elements.

We can talk about groups in terms of generators and relations. For D_{2n}, we write

$$D_{2n} = \langle \sigma, \tau : \sigma^n = \tau^2 = 1, \tau\sigma\tau = \sigma^{-1} \rangle$$

There's an explicit embedding $D_{2n} \to S_n$ as we can see above, but we can also think about D_{2n} in abstract.

The last type of group to know is the alternating group $A_n \subset S_n$. This contains exactly half the elements, and can be described as the kernel of the map

$$\operatorname{sgn}\colon S_n \to C_2 = \{\pm 1\}$$

which sends an m_1-m_2-$\cdots$-m_k-cycle to $(-1)^{m_1+\cdots+m_k-k}$. The alternating group consists of the identity element, m-cycles for odd m, 2-2-cycles, etc. There's another description of this that's not worth getting into right now.

7.3. **Abelian groups.** We now need to state the fundamental theorem on finitely generated abelian groups. Finitely generated is pretty obvious to define, but what's the theorem?

Theorem 7.15 (FTFGAG)**.** Let A be a finitely generated abelian group. Then

$$A \cong \mathbb{Z}^r \times \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_1 \mid n_2 \mid \cdots \mid n_k$. Alternatively,

$$A \cong \mathbb{Z}^r \times \mathbb{Z}/p_1^{\alpha_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_\ell^{\alpha_\ell}\mathbb{Z}$$

for primes p_i and powers α_i.

Now, what the hell does any of this mean? $\mathbb{Z}/n\mathbb{Z}$ is the cyclic group C_n, but we think of it additively and in terms of modular arithmetic. In particular, it's the quotient of $\mathbb{Z}$ by the normal subgroup $n\mathbb{Z} = \{n \cdot m : m \in \mathbb{Z}\}$, which we think of as generated under addition by 1. In an abelian group, all subgroups are normal, so there's no problem there. The product is the same as the product of sets, and the group operation works in the obvious way.

Problem 7.16. Let $m, n \in \mathbb{N}$. Then $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/mn\mathbb{Z}$ if and only if $(m, n) = 1$.

You can check both directions of that if you want. That means that the product of cyclic groups is still cyclic if and only if each of the terms are pairwise coprime. The two ways we think about the above decomposition depend how we group up the prime factors. We can either separate them as much as possible, or we can group them together. The specific details don't matter too much, but remember that theorem well.

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7.4. **Rings.** Now we can talk about rings.

Definition 7.17. A (unital) ring is a set R such that $(R, +)$ is an abelian group with another operation $\cdot$ satisfying:

- Multiplication is associative
- There exists $1 \in R$ such that $1 \cdot a = a \cdot 1 = a$ for all $a \in R$
- The distributive property holds: $a \cdot (b+c) = a \cdot b + a \cdot c$ and $(b+c) \cdot d = b \cdot d + c \cdot d$ for all $a, b, c, d \in R$

The multiplication is not necessarily commutative, as in the ring of matrices $M_n(F)$. Here's something that ought to be true and is:

Problem 7.18. Prove that $0 \cdot a = 0$ for all $a \in R$.

There are again two types of substructures that need to consider.

Definition 7.19. A subring $S \subset R$ is an abelian subgroup that is closed under multiplication. Sometimes we demand that $1 \in S$, sometimes we don't.

Subrings aren't even that important.

Definition 7.20. A left ideal $I \subset R$ is an abelian subgroup I that is closed under left multiplication: for every $a \in R$ and $x \in I$, $a \cdot x \in I$. Similarly, we can define a right ideal and a two-sided ideal.

Ideals are important, subrings aren't. Here's another definition and an important consequence.

Definition 7.21. An element $a \in R$ is invertible if there exists $b \in A$ such that $a \cdot b = b \cdot a = 1$.

Problem 7.22. If $I \subset R$ is an ideal and $a \in I$ is invertible, then $I = R$.

This means that in a proper ideal (i.e. $I \neq R$), we can't have any invertible elements.

There's one more type of element that needs defining:

Definition 7.23. An element $a \in R$ is a zero divisor if there exists $b \in R$ such that $a \cdot b = 0$.

Note that a zero divisor can't be invertible!

Ideals are also closed under intersection, sums, and not quite under products. For products we also have to take finite sums:

$$IJ := \left\{ \sum_{k=1}^{n} i_k j_k : i_k \in I, j_k \in J \right\}$$

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We can also talk about the left, right, or two-sided ideal generated by a subset of R. Finally, we can prove that if $I \subset R$ is a two-sided ideal, then R/I has the structure of a ring.

Now, what are the functions?

Definition 7.24. A (unital) ring homomorphism $\varphi\colon R \to S$ is an abelian group homomorphism such that $\varphi(1_R) = 1_S$ and $\varphi(r_1 \cdot r_2) = \varphi(r_1) \cdot \varphi(r_2)$.

As an example, consider the map $\varphi\colon \mathbb{Z} \to \mathbb{Z}$ such that $\varphi(n) = -n$. This is a perfectly good abelian group homomorphism, but it's not a ring homomorphism. In fact, since we demand that $\varphi(1) = 1$, there is only ever one map $\varphi\colon \mathbb{Z} \to R$ for any ring R, and it's defined by $\varphi(1) = 1_R$.

Kernels of ring homomorphisms are two-sided ideals, which is convenient, so that $R/\ker\varphi$ has the structure of a ring.

7.5. **Modular arithmetic.** Let $R = \mathbb{Z}$ and let $I = n\mathbb{Z}$. Then in the ring $\mathbb{Z}/n\mathbb{Z}$, we can do mathematics. The key is that we are working with 'remainders after dividing by n'. Let $n = 12$. Then for two examples,

$$8 + 7 = 15 \equiv 3, \quad 4 \cdot 5 = 20 \equiv 8$$

We can identify what elements in $\mathbb{Z}/n\mathbb{Z}$ are invertible and which are zero divisors. Supposing that d is a divisor of n, we know that $d \cdot n/d = n \equiv 0$. Even if $(d, n) = \alpha > 1$, it is still a zero divisor, because $d \cdot n/\alpha \equiv 0$. On the other hand, if $(d, n) = 1$, then we know that there's a solution to the expression $\alpha \cdot d + \beta \cdot n = 1$, so that $\alpha \cdot d \equiv 1$ and d is invertible.

7.6. **Fields.** There's a special situation that we can see immediately. Suppose that $n = p$ is a prime. Then every $d \in \mathbb{Z}/p\mathbb{Z}$ is coprime to p, so that every element of $\mathbb{Z}/p\mathbb{Z}$ is invertible. A commutative ring in which every element is invertible is called a field.

But wait, we already know a lot of fields. We know $\mathbb{Q}, \mathbb{R}, \mathbb{C}$, and others! Well surprise, there are also finite fields. Field homomorphisms are just ring homomorphisms, but with a twist: let's examine $\varphi\colon F \to K$ for two fields F, K. We know that $\ker\varphi \subset F$ is a two-sided ideal. But since every element in F is invertible, we know that either $\ker\varphi = F$ or $\ker\varphi = \{0\}$. Since $\varphi(1_F) = 1_K$, we know that $\ker\varphi$ can't be everything. Thus $\ker\varphi = \{0\}$ and all field homomorphisms are injective.

A special case is that of field automorphisms. It's pretty hard to find field automorphisms sometimes. This is the realm of Galois theory, which isn't particularly covered on the GRE. As a special observation, there are no nontrivial automorphisms of $\mathbb{Q}$ or $\mathbb{F}_p$ (which is $\mathbb{Z}/p\mathbb{Z}$ when it has its field clothes on).

Problem 7.25. Prove that if $\varphi\colon \mathbb{Q} \to \mathbb{Q}$ with $\varphi(0) = 0$ and $\varphi(1) = 1$, then $\varphi = \mathrm{id}_{\mathbb{Q}}$.

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8. Day 8: Analysis and topology

Topics covered: Lipschitz and uniform continuity of functions, absolute and uniform convergence of functions, suprema and infema. The main topology topics are: compactness, connectedness and path connectedness, continuous functions, metrics and metric spaces, separations axioms (e.g. Hausdorff), base of a topology.

8.1. A few basics.

Let's go over some of the fancy analysis words when it comes to sequences in $\mathbb{R}$.

Definition 8.1. Let $S \subset \mathbb{R}$ be any subset. Then we define the supremum $\sup S$ to be the number $\alpha \in \mathbb{R} \cup \{\infty\}$ such that $\alpha > s$ for all $s \in S$ and, if $\beta < \alpha$, then there exists $s' \in S$ such that $\beta < s'$.

We can similarly define the infimum of S by $\inf S = -\sup(-S)$. The supremum is also called the least upper bound and the infimum the greatest lower bound.

There's a nice feature about the real numbers that's related to its completeness (see below).

Theorem 8.2. Every bounded above subset of the real numbers has a supremum.

As another random note, here's a theorem.

Theorem 8.3. Suppose that $\{x_n\}$ is an increasing sequence, i.e. $x_n \leq x_{n+1}$ for all n. If the set of values $S = \{x_n\}$ is bounded above, then

$$\lim_{n\to\infty} x_n = \sup S.$$

8.2. Metric spaces.

Everything we say is going to work in an arbitrary (complete) metric space, so let's go ahead and do that definition first.

Definition 8.4. Let X be a set. A metric on X is a function $d\colon X \times X \to \mathbb{R}^{\geq 0}$ such that

- $d(x, y) = 0$ if and only if $x = y$
- $d(x, y) = d(y, x)$ for all $x, y \in X$
- For all $x, y, z \in X$, we have $d(x, y) + d(y, z) \geq d(x, z)$ (triangle inequality)

In any metric space, we obtain a topology by defining open sets to be generated the balls $B(\varepsilon, x)$ for every $\varepsilon \in \mathbb{R}$ and $x \in X$. Technically this is called the base of a topology. We'll go more into that later. That said, the definition of continuity for functions $\mathbb{R}^n \to \mathbb{R}^m$ can be repeated verbatim for functions $f\colon X \to Y$ between metric spaces, so we'll give that generality.

Definition 8.5. We say a function $f\colon X \to Y$ is uniformly continuous if for all $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $d_X(x_1, x_2) < \delta$, $d_Y(f(x_1), f(x_2)) < \varepsilon$. You may fill in $X \subset \mathbb{R}$ and $Y = \mathbb{R}$ if that makes you happier.

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This is strictly stronger than continuous: it says that the same δ can be used at any point in the domain, not just at the particular point a you want. That's why we don't talk about '$f(x)$ is uniformly continuous at $x = a$'.

There's an upgrade to this whose definition bears mentioning.

Definition 8.6. A function $f\colon X \to Y$ is called Lipschitz continuous if there exists a constant $K > 0$ such that $d_Y(f(x_1), f(x_2)) \leq K \cdot d_X(x_1, x_2)$ for all $x_1, x_2 \in X$.

In particular, the choice $\delta = \varepsilon/K$ shows that Lipschitz continuous implies uniformly continuous (implies continuous).

There's a nice way to conclude a function is uniformly continuous.

Problem 8.7 (Heine-Cantor Theorem)**.** If X is a compact metric space and $f\colon X \to Y$ is continuous, then it is also uniformly continuous.

Solution. Recall that one version of compact (we'll re-recall it later) is that every open cover of X admits a finite subcover. Since X is continuous, let's define the sets U_x for all $x \in X$ as follows:

$$U_x = \{x' \in X : d_Y(f(x), f(x')) < \varepsilon/2\}$$

In other words, it's the set around x that satisfy the conditions of uniform continuity for a slightly smaller epsilon. We can then define B_x to be the biggest open ball $B(\delta_x, x) \subset U_x$. For one more refinement, consider $B'_x = B(\delta_x/2, x)$, the ball with half the maximal radius. The collection $\{B'_x\}$ is (obviously) an open cover of X, so there's some finite collection $x_1, \ldots, x_n$ such that $B'_{x_i} = B(\delta_i/2, x_i)$ cover X.

Consider now $\delta = \dfrac{1}{2} \min \delta_i$. This is a positive number because we are taking a minimum (instead of, say, an infimum). Moreover, take any $z_1, z_2 \in X$ with $d_X(z_1, z_2) < \delta$. Without loss of generality, we have $z_1 \in B'_1$. Then

$$d(z_2, x_1) \leq d(z_2, z_1) + d(z_1, x_1) < \delta + \frac{\delta_1}{2} \leq \frac{\delta_1}{2} + \frac{\delta_1}{2} = \delta_1$$

which implies that $z_1, z_2 \in B_1$.

$$d_Y(f(z_1), f(z_2)) \leq d_Y(f(z_1), f(x_1)) + d_Y(f(x_1), f(z_2)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

which proves that $f(x)$ is uniformly continuous.

This proof is a good reminder of the utility of compactness, and that on compact domains many results are upgradeable from local (i.e. continuity begin defined at points) to global (having a uniform δ for each ε).

Uniformly continuous functions have a nice property.

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Theorem 8.8. Suppose that $f\colon Z \subset X \to Y$ is uniformly continuous on a subset $Z \subset X$. Then there is a unique extension $\overline{f}\colon \overline{Z} \to Y$ defined on the closure of Z that is still continuous.

This doesn't work if the function isn't uniformly continuous. For instance, let $f(x) = 1/x$ be defined on $f\colon (0,1) \to \mathbb{R}$. Then there is no way to continuously extend $f(x)$ to $[0,1]$ as we would need

$$f(0) = \lim_{x\to 0} \frac{1}{x}$$

and this limit diverges. The case of uniform continuity ensures that we don't run into this problem.

There's also absolute continuity, but I don't think we need to recall that.

8.3. **Convergence of functions.** We can now begin to talk about convergence of functions. We will restrict our attention to $Y = \mathbb{R}$ and $X \subset \mathbb{R}$ because we will care about completeness. Let's recall that briefly.

Definition 8.9. A sequence $\{x_n\}$ in a metric space X is called Cauchy if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $d_X(x_m, x_n) < \varepsilon$ whenever $n, m > N$.

The terms of a Cauchy sequence get arbitrarily close together. This is related to the sequence converging to some limit.

Problem 8.10. Prove that if $\lim_{n\to\infty} x_n = x$ exists, then $\{x_n\}$ is Cauchy.

The converse is not necessarily true. Consider the metric space $\mathbb{Q}$ and the sequence

$$1, 1.4, 1.41, 1.414, \ldots$$

given by the truncations of the infinite decimal $\sqrt{2}$. The limit is, by design, not in $\mathbb{Q}$, however this sequence is Cauchy as $|x_n - x_{n+1}| < 10^{-n+1}$ so gets arbitrarily small. We therefore get our definition:

Definition 8.11. A metric space X is called complete if every Cauchy sequence converges to some limit in X.

It'll be convenient to work in complete metric spaces so that sequences are Cauchy if and only if they are convergent.

Definition 8.12. Consider a sequence of functions $f_n\colon X \to \mathbb{R}$. Then we can define a new function $f\colon X \to \mathbb{R}$ by

$$f(x) = \lim_{n\to\infty} f_n(x)$$

assuming all these limits exist. In this case, we say that $\{f_n\}$ converges to f pointwise.

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This is pretty good. For instance, let $f_n\colon [0,1] \to \mathbb{R}$ be defined by $f_n(x) = x^n$. Then it's pretty clear that the pointwise limit exists and

$$f(x) = \begin{cases} 0 & x \in [0,1) \\ 1 & x = 1 \end{cases}$$

This presents our conundrum. Each of the functions $f_n(x)$ is continuous, but their pointwise limit is not! We need to introduce more refined version of convergence that takes into account that we have an entire function, not just a series of points.

Definition 8.13. Let $\{f_n\colon X \to \mathbb{R}\}$ be a sequence of functions. We say that $\{f_n\}$ converge to f uniformly if they converge pointwise and, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|f_n(x) - f(x)| < \varepsilon \text{ for all } n > N$$

That is, the pointwise limits are all getting close to the limit function $f(x)$ simultaneously.

We can see how this should be generalised to general metric spaces. Note that since $\mathbb{R}$ is complete, we could also demand that the sequence $\{f_n\}$ is uniformly Cauchy rather than uniformly continuous, which is sometimes easier.

The point is this:

Theorem 8.14. Suppose that $\{f_n\}$ is a sequence of continuous functions that converge uniformly to f. Then f is also continuous.

This must mean that $f_n(x) = x^n$ does not converge uniformly to the limit function. To see this, fix $1 > \varepsilon > 0$ and any $N \in \mathbb{N}$. We will show that there exists $x \in [0,1]$ such that $|f_N(x) - f(x)| > \varepsilon$. Specifically, we are going to choose an $x \in (0,1)$ so we just need to prove that $x^N > \varepsilon$. But this is easy: take any $1 > \delta > \varepsilon$ and let $x = \delta^{1/N}$.

Can we upgrade this theorem in the case that we know that $\{f_n\}$ are also uniformly continuous? Yes. Leave it at that.

8.4. **Integrals.** Now let's address the issue of integrals. Suppose that we have a sequence of integrable functions $f_n\colon X \to \mathbb{R}$ where, again, we will consider $X \subset \mathbb{R}^n$ (and most likely $n = 1$). Suppose further that $f_n \to f$ pointwise. Does it follow that f is integrable? In particular, do we have an equality

$$\lim_{n\to\infty} \int_X f_n(x)\,dx = \int_X f(x)\,dx$$

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The answer is not necessarily. An easy counterexample is the following: let $f_n\colon (0,1] \to \mathbb{R}$ be defined by

$$f_n(x) = \begin{cases} n & x \in (0, 1/n] \\ 0 & \text{else} \end{cases}$$

Then each $f_n(x)$ isn't continuous, but it is bounded with finitely many discontinuities, and

$$\int_0^1 f_n(x)\,dx = 1$$

for all $n \in \mathbb{N}$.

Now, $f_n \to 0$ pointwise because $f_n(x) = 0$ for all $n > 1/x$. But, as one observes

$$\int_0^1 0\,dx \neq 1$$

Thus, something has gone wrong. As something specific to notice, the functions $\{f_n\}$ are not uniformly bounded by any constant. Consequently, $\{f_n\}$ does not converge to the zero function uniformly. We have two theorems that give us the means to commute the integral and the limit.

Theorem 8.15 (Uniform convergence theorem)**.** If $f_n \to f$ uniformly as functions $[a,b] \to \mathbb{R}$ and all f_n are integrable, then

$$\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx$$

Theorem 8.16. Suppose $f_n \to f$ pointwise as functions $[a,b] \to \mathbb{R}$ and all f_n are integrable. Suppose further that for all $n \in \mathbb{N}$, $|f_n(x)| \leq g(x)$ for an integrable function $g\colon [a,b] \to \mathbb{R}$. Then

$$\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx$$

I'm trying to avoid measure theory in the description of this theorem, and I think we don't need it.

8.5. **Topology.** I think we can now talk about topology in abstract.

Definition 8.17. A topological space is a set X along with a subset $\mathcal{T} \subset \mathcal{P}(X)$ such that:

- $\varnothing, X \in \mathcal{T}$
- If $\{U_i\}_{i\in I}$ is a collection of elements in $\mathcal{T}$, then so is $\bigcup_{i\in I} U_i$

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- If $U_1, \ldots, U_n$ is a finite collection of elements in $\mathcal{T}$, then so is $\bigcap_{i=1}^{n} U_i$

The set $\mathcal{T}$ is called a topology on X. The sets $U \in \mathcal{T}$ are called open sets. A set V such that $V^c \in \mathcal{T}$ is called closed. Note that you could also define a topology using closed sets and a dual set of axioms.

Every subset $S \subset X$ has an interior and a closure. The interior S° is the union of all open sets $U \subset S$ and the closure $\overline{S}$ is the intersection of all closed sets $S \subset C$.

There are always two topologies on any set X, namely the maximal choice $\mathcal{T} = \mathcal{P}(X)$ called the discrete topology and the minimal choice $\{\varnothing, X\}$ called the indiscrete topology (which I think is joke).

Something that we might care about is when two topologies are the same, i.e. when they have exactly the same open sets. Well, usually a topology is defined using a generating set, in the following sense:

Definition 8.18. A subset $B \subset \mathcal{T}$ is called a base of the topology $\mathcal{T}$ on X if:

- $\bigcup_{U \in B} U = X$
- For every $U_1, U_2 \in B$ and every $x \in U_1 \cap U_2$, there exists a $U_3 \in B$ containing x

Problem 8.19. Suppose that B_1 is a base of $\mathcal{T}_1$ and B_2 a base of $\mathcal{T}_2$ on a set X. Suppose further that for every $U_2 \in B_2$, there exists $U_1 \in B_1$ such that $U_1 \subset U_2$ and vice versa. Then $\mathcal{T}_1 = \mathcal{T}_2$.

If we have that $\mathcal{T}_1 \subset \mathcal{T}_2$, we say that $\mathcal{T}_1$ is coarser than $\mathcal{T}_2$, or that $\mathcal{T}_2$ is finer than $\mathcal{T}_1$. In linguistic terms, having more open sets makes the set X more smooth. We can also check this on bases.

8.6. **Separation axioms.** There is an increasing list of axioms that make topological spaces more and more nice. We'll give a list and examples.

A topological space X is T_0 if for every two points $x, y \in X$, there exists an open set U such that $x \in U$ but $y \notin U$. That is, all points are topologically distinguishable.

A topological space X is T_1 if for every two points $x, y \in X$, there exists an open set U such that $x \in U$ but $y \notin U$ and an open set V such that $y \in V$ and $x \notin V$. In this case, points are closed.

A topological space X is T_2 or Hausdorff if for every two points $x, y \in X$, there exist open sets U, V such that $x \in U$, $y \in V$ and $U \cap V = \varnothing$. That is, points are separated.

A topological space X is regular if for every $x \in X$ and closed $K \subset X$ such that $x \notin K$, there exist open sets U, V such that $x \in U$, $K \subset V$, and $U \cap V = \varnothing$. That is, points and closed sets are separated. X is T_3 (or regular Hausdorff) if it is regular and T_0.

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A topological space X is normal if for every $C, K \subset X$ closed with $C \cap K = \varnothing$, there exist U, V open such that $C \subset U$, $K \subset V$, and $U \cap V = \varnothing$. That is, closed sets are separated.

On this note,

Theorem 8.20 (Urysohn's lemma)**.** A topological space X is normal if and only if for any disjoint closed sets $C, K \subset X$, there exists a continuous function $f\colon X \to [0,1]$ such that $f(C) = 0$ and $f(K) = 1$. That is, C and K are separated by a continuous function.

A topological space is T_4 (or normal Hausdorff) if it is T_1 and normal. All metric spaces are T_4. There's another theorem that's worth stating too.

Theorem 8.21 (Urysohn's theorem)**.** Let X be a topological space. Then X is separable and metrizable (i.e. admits a metric that generates its topology) if and only if it is regular, Hausdorff, and second-countable.

We're missing some of these words. A topological space is separable if it admits a countable dense subset, i.e. there's a countable set $S \subset X$ such that $\overline{S} = X$. A topological space is second-countable if it admits a countable base.

8.7. **Continuity.** What are the functions we care about?

Definition 8.22. Let X, Y be two topological spaces. Then a set map $f\colon X \to Y$ is called continuous if $f^{-1}(V)$ is open for any open $V \subset Y$. Equivalently, if $f^{-1}(C)$ is closed for every closed $C \subset Y$.

Problem 8.23. Prove that if X, Y are metric spaces endowed with the metric topology, this is the same definition as the usual one.

Definition 8.24. A map $f\colon X \to Y$ is called a homeomorphism if it is continuous, bijective, and moreover sends open sets to open sets.

Without this last condition, there's no guarantee that the set-theoretic inverse is a continuous function.

Now, what kind of sets are there besides open and closed?

Definition 8.25. A set $Z \subset X$ is called disconnected if there exist open sets $U, V \subset X$ such that $U \cup V = Z$ and $U \cap V = \varnothing$. A set that is not disconnected is called connected.

Definition 8.26. A set $Z \subset X$ is called path-connected if for every $a, b \in Z$, there exists a continuous function $\gamma\colon [0,1] \to Z$ such that $\gamma(0) = a$ and $\gamma(1) = b$.

Problem 8.27. Prove that every path-connected set is connected.

The converse is not true.

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Problem 8.28. Prove that the graph of the function

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

is connected but not path connected.

Definition 8.29. A set $K \subset X$ is called compact if every open cover of K admits a finite subcover.

Compact sets are necessarily closed and bounded. If X is Euclidean space, we get the converse:

Theorem 8.30 (Heine-Borel Theorem)**.** A set $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.

There are a couple extra things one should prove now.

Problem 8.31. The image of a connected set under a continuous function is connected. The image of a compact set under a continuous function is compact.

I think that's about everything.

9. Day 9: Miscellaneous

Topics covered: probability and combinatorics, statistics, geometry, set theory, logic, graph theory, algorithms.

9.1. **Combinatorics.** Pigeonhole principle (go over it again)

Permutations and combinations (and factorials)

Specific case: going in a circle, it's $(n-1)!$ instead of $n!$ because you have to divide by n options.

9.2. **Probability via area.** We can imagine a probability space Ω and events A and B being subsets of Ω, such that the area/volume of Ω is 1 and hence $P(A)$ is given by the volume or area of A. Then looking at the probability of $P(A \text{ and } B)$ is just given by the intersection of these areas, and similarly for $P(A \text{ or } B)$.

When you're trying to compare continuous random variables $x, y, z \in [0, 1]$ (for instance), the volume approach is very useful, as we've seen.

Problem 9.1. If x, y are randomly chosen in $[0, 1]$, what is the probability that $x \geq 2y$?

In Progress

Solution. We can picture this as the double integral where $y \in [0,1]$ and $x \in [2y, 1]$. Except that this doesn't make total sense, because $2y > 1$ when $y > 1/2$, so we really have to integrate $y \in [0, 1/2]$.

$$\int_0^{1/2} \int_{2y}^{1} 1 \, dx \, dy = \int_0^{1/2} 1 - 2y \, dy = \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

We can also do this via drawing the picture and computing the area of the triangle.

The same works in 3d.

9.3. **General probability.** Conditional probability: $P(A|B) = \dfrac{P(A \text{ and } B)}{P(B)}$. Read 'probability of A given B.

We say that A and B are independent if $P(A \text{ and } B) = P(A) \cdot P(B)$. Equivalently, $P(A|B) = P(A)$.

There's a nice way to swap the order of conditional probability, called Bayes' theorem.

Theorem 9.2.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

In action:

Problem 9.3. Consider drawing two cards from a deck. Compute the probability of drawing a spade first given that you drew a spade second.

Solution. Let A be the first spade and B the second spade. We can work out the probability $P(B|A)$ explicitly: there are 51 cards left and 12 spades, so $P(B|A) = 12/51$. We can also compute $P(A)$ and $P(B)$: $P(A) = P(B) = 1/4$. We can see that $P(B) = 1/4$ by noting that it's just taking a random card from the deck. Thus $P(A|B) = P(B|A) = 12/51$.

Now these have been discrete probabilities here, but what about continuous ones?

Definition 9.4. A probability distribution function for a random variable X is a positive integrable function $f \colon \mathbb{R} \to \mathbb{R}$ such that

$$\int_{-\infty}^{\infty} f(x) \, dx = 1$$

It has a corresponding continuous distribution function defined by

$$P(X \leq a) = F(a) = \int_{-\infty}^{a} f(x) \, dx$$

In Progress

9.4. **Statistics.** The expected value of a discrete random variable X (taking values in $\mathbb{R}$) is

$$E(X) = \sum_{A \in X} P(A) \cdot A$$

For a continuous random variable with pdf $f(x)$ is

$$E(X) = \int_{\infty}^{\infty} x \cdot f(x)\, dx$$

The variance can be calculated as $E(X^2) - E(X)^2$. Specifically,

$$\sum_{A \in X} P(A)(A - E(X))^2, \quad \int_{-\infty}^{\infty} (x - E(X))^2 \cdot f(x)\, dx$$

The standard deviation σ is the square root of the variance.

What do we know about standard deviation? Well, suppose we have a normal distribution. That's basically none of the above examples, but we'll get to that soon. Within ± 1 standard deviation of the expected value (or the mean) is 68% of the distribution, within $\pm 2\sigma$ is 95% and within $\pm 3\sigma$ is 99.7%.

When can we expect a variable to be normally distributed? For example, Bernoulli trials. Suppose we have an event with probability p and we perform n trials. Then the expected value of successful trials is $n \cdot p$. If we repeat this situation a bunch of times, we can look at the number of trials that were actually successful. The variance of this distribution is $n \cdot p \cdot (1 - p)$ and so the standard deviation is the square root of this.

Problem 9.5. Suppose we roll a 20-sided die 400 times. Consider the probability of rolling a prime number. What is the expected number of successes and what is the standard deviation?

Solution. $p = 7/20$ so go from there.

9.5. **Geometry.** Triangles: let's start there. There's the law of sines and the law of cosines, which I can recall, but there's also Heron's formula for the area of a triangle: if the sides are a, b, c, then

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

where $s = \dfrac{a+b+c}{2}$ is the semiperimeter.

This is pretty nice when you know that the triangle is equilateral and then the area becomes $\dfrac{x^2\sqrt{3}}{4}$ where x is the side length.

The measure of the angle of a regular n-gon is $\dfrac{180(n-2)}{n}$. That can be useful.

In Progress

You can also try to find the area of an inscribed polygon or a circumscribed polygon around a circle. We can do the example of a hexagon and see where it goes from there.

Problem 9.6. Do that, I think I can wing it. Start with the unit circle and go from there.

9.6. **Set theory.** We've discussed the issue of cardinality. If you take the power set, the cardinality strictly goes up. We can do some discussion of countability though.

$\aleph_0$ is the cardinality of $\mathbb{N}$. We say that a set X is countable if there exists a surjective function $\mathbb{N} \to X$ or an injective function $X \to \mathbb{N}$. A countable union of countable sets is still countable, and a finite product of countable sets is countable, but a *countable* product of countable sets is definitely not countable anymore.

In particular, consider the set $X = \{0, \ldots, 9\}$ and take an infinite product $\prod_{\mathbb{Z}} X$. Then if we interpret the elements of this product as $x_n \cdot 10^n$ for $n \in \mathbb{Z}$, then we get (roughly) $\mathbb{R}$ as long as we make sure that these n are bounded above, but not necessarily bounded below. We just have to throw away the elements that aren't bounded above, but that collection is finite. Hence uncountable minus countable gives us an uncountable set $\mathbb{R}$. Great!

9.7. **Graph theory.** Graphs are made of edges and vertices. A cycle is a cycle. Sometimes graphs are directed, sometimes they aren't. I guess that's about it.

9.8. **Algorithms.** Learn some Python? If you don't know any computer science, it's a bit tricky. I guess just try to treat the algorithm like a proof with input.

In Progress